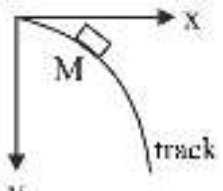
PHYSICS

1) A man M slides down a curved frictionless track, starting from rest. The curve obeys the equation $y = \frac{x^2}{2}$. The tangential acceleration of man is



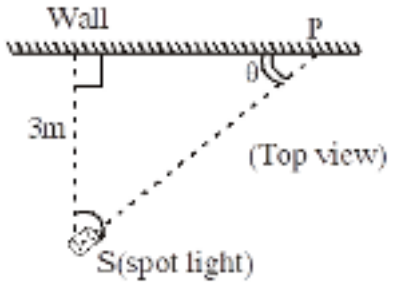
(1) g

(2) $\frac{gx}{\sqrt{x^2+4}}$

(3) $\frac{g}{2}$

(4) $\frac{gx}{\sqrt{x^2+1}}$

2) A spot light S rotates in a horizontal plane with a constant angular velocity of 0.1 rad/s. The spot of light P moves along the wall at a distance 3m. What is the velocity of the spot P when $\theta = 45°$?



(1) 0.6 m/s

(2) 0.5 m/s

(3) 0.4 m/s

(4) 0.3 m/s

3) One end of a string of length $\ell = \frac{14}{9}$m is fixed and a mass of 1 kg is tied to the other end. The ball is given a velocity $2\sqrt{g\ell}$ at the bottom most point A as shown in figure. The string is cut when it becomes horizontal. Find the total distance travelled by ball till it stop for the 1st time (Take $\pi = \frac{22}{7}$



).

(1) 3 m

(2) 28/9 m

(3) 18/7 m

(4) 4 m

4) A grind-stone starts revolving from rest, if its angular acceleration is 4.0 rad/sec^2 (uniform) then after 4 sec. What is its angular displacement & angular velocity respectively:

(1) 32 rad, 16 rad/sec

(2) 16 rad, 32 rad/sec

(3) 64 rad, 32 rad/sec

(4) 32 rad, 64 rad/sec

5) A motorcyclist moving with a velocity of 72 km per hour on a flat road takes a turn on the road at a point, where the radius of curvature of the road is 20 meter. The acceleration due to gravity is 10 m/sec^2. In order to avoid skidding, he must not bend with respect to the vertical by an angle greater than :

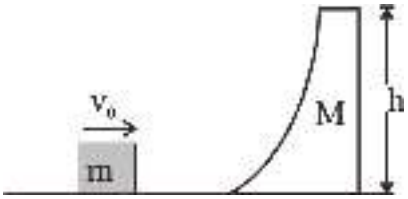
(1) $\theta = \tan^{-1} 6$

(2) $\theta = \tan^{-1} 2$

(3) $\theta = \tan^{-1} (25.92)$

(4) $\theta = \tan^{-1} 4$

6) A block of mass m is pushed towards the movable wedge of mass M and height h, with a velocity v_0. All surfaces are smooth. The minimum value of v_0 for which the block will reach the top of the wedge is :



(1) $\sqrt{2gh}$

(2) $\sqrt{\frac{2ghm}{M}}$

(3) $\sqrt{\frac{2gh(m+M)}{M}}$

(4) $\sqrt{\frac{2ghM}{(m+M)}}$

7) A body of mass m moving with a velocity u in the +x-direction collides with a body of mass M moving with velocity V in the +y-direction. They stick together during collision. Then :

(1) the loss of kinetic energy as a result of collision is $\frac{1}{2}\left(\frac{Mm}{M+m}\right)(V^2+u^2)$

(2) the composite body moves in a direction making an angle $\theta = \tan^{-1}\left(\frac{MV}{mu}\right)$ with the x-axis

(3) the magnitude of momentum of the composite body is $\sqrt{(mu)^2+(MV)^2}$

(4) All the above choice are correct

8) A wooden block of mass *M* rests on a horizontal surface. A bullet of mass *m* moving in the horizontal direction strikes and gets embedded in it. The combined system covers a distance *x* on the surface. If the coefficient of friction between wood and the surface is μ, the speed of the bullet at the time of striking the block is (where *m* is mass of the bullet)

(1) $\sqrt{\frac{2Mg}{\mu m}}$

(2) $\sqrt{\frac{2\mu mg}{Mx}}$

(3) $\sqrt{2\mu gx}\left(\frac{M+m}{m}\right)$

(4) $\sqrt{\frac{2\mu mx}{M+m}}$

9) A cart of mass M is tied to one end of a massless rope of length 10m. The other end of the rope is in the hands of a man of mass $\frac{M}{2}$. The entire system is placed on a smooth horizontal surface. If the man pulls the cart by the rope then find the distance the man will slip on the horizontal surface before the man & the cart will meet each other.

(1) $\frac{20}{3}$m

(2) $\frac{10}{3}$m

(3) 0

(4) None

10) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R),

Assertion (A) : When a fire cracker (rocket) explodes in mid air, its fragments fly in such a way that they continue moving in the same path, which the fire cracker would have followed, had it not exploded

Reason (R) : Explosion of cracker (rocket) occurs due to internal forces only and no external force

acts for this explosion.
In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(3) **(A)** is correct but **(R)** is not correct

(4) **(A)** is not correct but **(R)** is correct

11) The rate of flow of water in a capillary tube of length and radius r is V. The rate of flow in another capillary tube of length 2 and radius 2r for same pressure difference would be :-

(1) 16 V

(2) 9 V

(3) 8 V

(4) 2 V

12) A block of silver of mass 4 kg hanging from a string is immersed in a liquid of relative density 0.75. If relative density of silver is 10, then tension (in N) in the string will be :
[take g = 10 m/s^2]

(1) 35

(2) 37

(3) 39

(4) 40

13) A metallic wire of density d is lying horizontal on the surface of water (surface tension = T). The maximum length of wire so that it may not sink will be :-

(1) $\sqrt{\frac{2Tg}{\pi d}}$

(2) $\sqrt{\frac{2\pi T}{dg}}$

(3) $\sqrt{\frac{2T}{\pi dg}}$

(4) any length

14) When the load on a wire is increased from 3 kg-wt to 5 kg-wt the elongation increases from 0.61 mm to 1.02 mm. The required work done during the extension of the wire is :-

(1) 16×10^{-3} J

(2) 8×10^{-2} J

(3) 20×10^{-2} J

(4) 11×10^{-3} J

15) An iron casting containing a number of cavities weighs W in air and W_0 in water. What is the

volume of cavities in the casting ? (density of iron & water are ρ_i & ρ_w respectively)

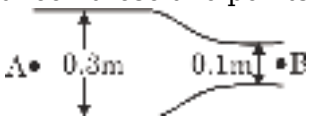
(1) $\frac{W}{g}\left[\frac{1}{\rho w}+\frac{1}{\rho_i}\right]-\frac{W_0}{g\rho w}$

(2) $\frac{W}{g}\left[\frac{1}{\rho w}+\frac{1}{\rho_i}\right]+\frac{W_0}{g\rho w}$

(3) $\frac{W}{g}\left[\frac{1}{\rho w}-\frac{1}{\rho_i}\right]+\frac{W_0}{g\rho w}$

(4) $\frac{W}{g}\left[\frac{1}{\rho w}-\frac{1}{\rho_i}\right]-\frac{W_0}{g\rho w}$

16) Water is flowing through a horizontal tube according to the figure. Its diameter at two points are 0.3 m and 0.1 m respectively. Pressure difference between these two points is equal to 0.8 m of water column. Find rate of flow of water in the tube.



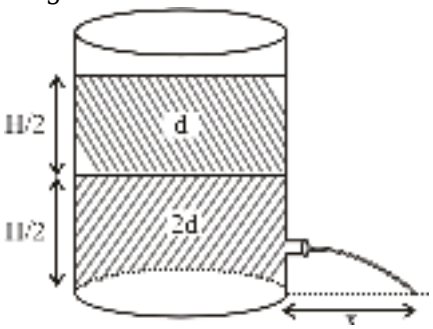
(1) 32 ltr/s

(2) 72 ltr/s

(3) 198 ltr/s

(4) 49.5 ltr/s

17) A container of large uniform cross-sectional area A is resting on a horizontal surface holds two immiscible, non-viscous and incompressible liquids of densities d and 2d each of height $\frac{H}{2}$ as shown. The lower density of liquid is open to atmosphere. A small hole is made on the wall of container at height $h\left(h<\frac{H}{2}\right)$ from floor. The initial speed of efflux of the liquid at the hole is :-



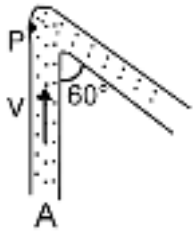
(1) (3H – 4h) g

(2) $\frac{(3H-4h)g}{2}$

(3) $\sqrt{(3H-4h)g}$

(4) $\sqrt{\frac{(3H-4h)g}{2}}$

18) Water (density ρ) is flowing through the uniform tube of cross-sectional area A with a constant speed v as shown in the figure. The magnitude of force exerted by the water on the curved corner of



the tube is (neglect viscous forces)

(1) $\sqrt{3}\rho Av^2$

(2) $2\rho Av^2$

(3) $\sqrt{2}\rho Av^2$

(4) $\frac{\rho Av^2}{\sqrt{2}}$

19) A plane is in level flight at constant speed and each of its two wings has an area of 25 m^2. If the speed of the air on the upper and lower surfaces of the wing are 270 km h^{-1} and 234 km h^{-1} respectively, then the mass of the plane is (Take the density of the air = 1 kg m^{-3})

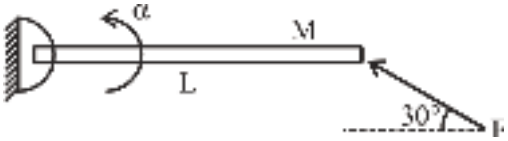
(1) 1550 kg

(2) 1750 kg

(3) 3500 kg

(4) 3200 kg

20) The thin rod shown below has mass M and length L. A force F acts at one end as shown and the rod is free to rotate about the other end in horizontal plane. Initial angular acceleration of the rod is :



(1) $\frac{3F}{2ML}$

(2) $\frac{2F}{3ML}$

(3) $\frac{F}{ML}$

(4) $\frac{F}{2ML}$

21) A metre stick is balanced on a knife edge at its centre. When two coins, each of mass 5 g are put one on top of the other at the 12.0 cm mark, the stick is found to be balanced at 45.0 cm. What is the mass of the metre stick ?

(1) 33g

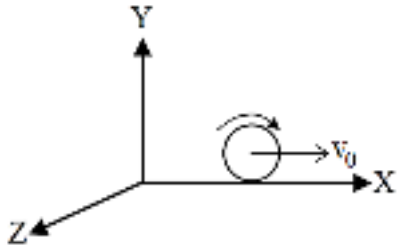
(2) 66g

(3) 99g

(4) None of these

22) A solid sphere rolls without slipping on a rough surface and the centre of mass has a constant

speed v_0. If the mass of the sphere is m and its radius is R, then find the angular momentum of the sphere about the point of contact.



(1) $\frac{7}{5} Mv^0R\left(-\hat{k}\right)$

(2) $\frac{3}{5} Mv_0R(-\hat{k})$

(3) $\frac{Mv_0R}{2}(-\hat{k})$

(4) $\frac{2}{5} Mv_0R(-\hat{k})$

23) **Statement I :** A body rolls down an inclined plane without slipping. The fraction of total energy associated with its rotation will depend on its radius of gyration.
Statement II : Total kinetic energy of rolling body is equal to addition of kinetic energy of rotation and kinetic energy of translation.

(1) Both statements are true

(2) Both statements are false

(3) Statement I is true and Statement II is false

(4) Statement I is false and Statement II is true

24) A solid sphere is rolling without slipping on a level surface at a constant speed of 2.0 ms^{-1}. How far can it roll up a 30° ramp before it stops? (g = 10 m/s^2)

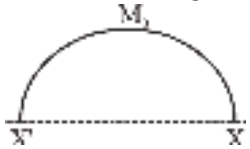
(1) 56 cm

(2) 26 cm

(3) 53 cm

(4) 84 cm

25) A rod of length L and mass M_0 is bent to form a semicircular ring as shown. The M.I. about X'X is

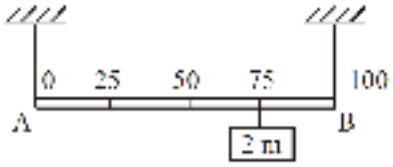
:- 

(1) $\frac{M_0L^2}{2\pi^2}$

(2) $\frac{M_0L^2}{\pi^2}$

(3) $\frac{M_0L^2}{4\pi^2}$

(4) $\frac{2M_0L^2}{\pi^2}$

26) 

Shown in the figure is rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass 'm' and has another weight of mass 2 m hung at a distance of 75 cm from A. The tension in the string at A is :

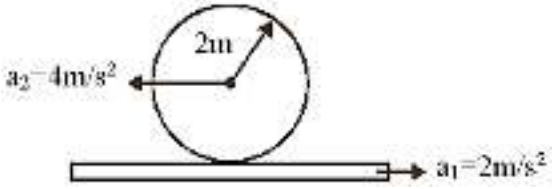
(1) 2mg

(2) mg/2

(3) (3/4)mg

(4) mg

27) In the given figure a sphere of radius 2m rolls without sliding on a plank. The acceleration of the sphere and plank are indicated. The value of angular acceleration α is :-



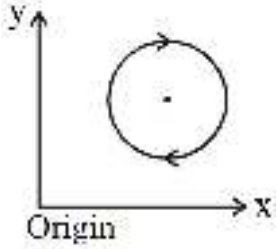
(1) 2 rad/s^2

(2) 1 rad/s^2

(3) 3 rad/s^2

(4) 4 rad/s^2

28) A particle is moving in circle with uniform speed as shown. The angular momentum of the particle w.r.t. origin is :-



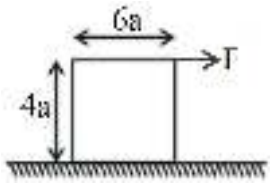
(1) Constant in magnitude only

(2) Constant in magnitude as well as direction

(3) Constant in direction only

(4) Variable in magnitude as well as direction

29) Minimum force F required to make the block topple about the edge is: (mass of block is m)



(1) 2mg/3

(2) 3mg/2

(3) 4mg/3

(4) 3mg/4

30) ***Assertion (A):*** The time period of revolution of a satellite close to surface of earth is smaller than that revolving far from surface of earth
Reason (R): The square of time period of revolution of a satellite is directly proportional to cube of its orbital radius.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

31) Five point masses m each are placed at five corners of a regular pentagon. Distance of any corner from centre is r. Match the following two columns.(Symbols have their usual meaning)

Column I		**Column II**	
A	Gravitational field strength at centre	P	Gm/r^2
B	Gravitational potential at centre	Q	4 Gm/r
C	When one mass is removed gravitational field strength (magnitude) at centre.	R	zero
D	When one mass is removed gravitational potential at centre	S	None of these

(1) A→R, B→S, C→P, D→S

(2) A→R, B→P, C→S, D→Q

(3) A→S, B→R, C→P, D→Q

(4) A→S, B→Q, C→R, D→P

32) Find gravitational field at point (2, 3) if gravitational potential is given by $V = 5x^2y$.

(1) 0

(2) $-80\hat{i}$

(3) $-60\hat{i}-20\hat{j}$

(4) $-60\hat{i}-80\hat{j}$

33) A projectile is projected with velocity kv_e in vertically upward direction from the surface of earth into the space (v_e is escape velocity and $k < 1$). If air resistance is considered to be negligible then the maximum height from the centre of earth to which it can go, will be: (R = radius of earth) :-

(1) $\frac{R}{k^2+1}$

(2) $\frac{R}{k^2-1}$

(3) $\frac{R}{1-k^2}$

(4) $\frac{R}{k+1}$

34) Two identical satellite A and B revolve round the earth in circular orbits at distance R and 3R from the surface of the earth. The ratio of the linear momentum of A and B is (R=radius of the earth) :-

(1) 1:1

(2) $1:\sqrt{2}$

(3) $\sqrt{2}:1$

(4) 2:1

35) The orbital velocity of an artificial satellite in a circular orbit very close to earth surface is V. The velocity of another satellite orbiting in circular orbit at height 3R from earth's surface will be (R = Radius of Earth) :-

(1) $\frac{V}{\sqrt{7}}$

(2) $\frac{V}{\sqrt{6}}$

(3) $\frac{V}{2}$

(4) $\frac{V}{\sqrt{2}}$

36) A ring of mass M and radius R is fixed in y–z plane with centre at origin. A point mass M_0 is placed at centre of ring. If point mass M_0 is displaced slightly along x-axis and released find the time when point mass again reach the centre.

(1) $\frac{\pi}{2}\sqrt{\frac{R^3}{GM}}$

(2) $\frac{\pi}{4}\sqrt{\frac{R^3}{GM}}$

(3) $2\pi\sqrt{\frac{R^3}{GM}}$

(4) $\pi\sqrt{\frac{R^3}{GM}}$

37) While revolving around the earth, a satellite has kinetic energy K, potential energy U and total

energy E. If the value of $\frac{K+U+E}{K-U}$ is x and $\frac{K-U-E}{E-K}$ is y then ratio $\frac{y}{x}$ is :-

(1) 2 : 3

(2) 2 : 1

(3) 3 : 1

(4) 3 : 2

38) A narrow glass U-tube whose one limb is of diameter 3.0mm and other limb of diameter 6.0 mm is partially filled with water. The difference in the level of water in two limbs will be
[Given density of water = 1.0×10^3 kg m^{-3}, surface tension of water = 7.2×10^{-2} Nm^{-1} ; angle of contact between glass and water to be taken as zero and g = 10ms^{-2}]

(1) 2.4mm

(2) 3.6mm

(3) 4.8mm

(4) zero

39) Match the terms in list-I with suitable options in list-II

	List-I		**List-II**
A.	Concave menisus	P	cohesive & adhesive forces
B.	Convex menisus	Q	cohesive forces
C.	Capillary action	R	adhesive forces dominate
D.	Bigger drop splits into smaller drops	S	temperature decreases
E.	Surface tension	T	cohesive forces dominate

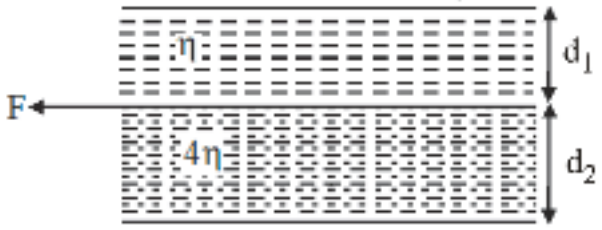
(1) A → R, B → T, C → S, D → P, E → Q

(2) A → T, B → R, C → P, D → Q, E → S

(3) A → R, B → T, C → P, D → S, E → Q

(4) A → P, B → S, C → R, D → T, E → Q

40) A thin plate separates two liquids of coefficients of viscosity η and 4η kept between two fixed plates as shown. If plate has to be pulled by applying minimum force then $\frac{d_2}{d_1}$ is :-



(1) 1

(2) 2

(3) 3

(4) 4

41) The angle of contact between glass and water is 0° and it rises in a capillary upto 6 cm when its surface tension is 70 dynes/cm. Another liquid of surface tension 140 dynes/cm, angle of contact 60° and relative density 2 will rise in the same capillary by:

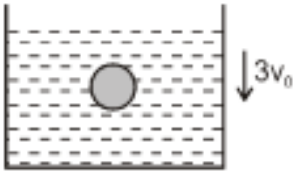
(1) 12 cm

(2) 24 cm

(3) 3 cm

(4) 6 cm

42) A container filled with viscous liquid is moving vertically downwards with constant speed $3v_0$. At the instant shown, a sphere of radius r is moving vertically downwards (in liquid) has speed v_0 wrt ground. The coefficient of viscosity is η. There is no relative motion between the liquid and the container. Then at the shown instant, the magnitude of viscous force acting on sphere is



(1) $6\pi\eta r v_0$

(2) $12\pi\eta r v_0$

(3) $18\pi\eta r v_0$

(4) $24\pi\eta r v_0$

43) A capillary tube with inner cross-section in the form of a square of side *a* is dipped vertically in a liquid of density ρ and surface tension σ liquid wet the surface of capillary tube with angle of contact θ. The approximate height to which liquid will be raised in the tube is : (Neglect the effect of surface tension at the corners of capillary tube)

(1) $\frac{2\sigma\cos\theta}{a\rho g}$

(2) $\frac{4\sigma\cos\theta}{a\rho g}$

(3) $\frac{8\sigma\cos\theta}{a\rho g}$

(4) None of these

44) The amount of force needed to pull out a regular hexagonal figure made of wire, perpendicular to its plane from liquid of surface tension 'T' will be:-
(assume mass of wire is m, side = a)

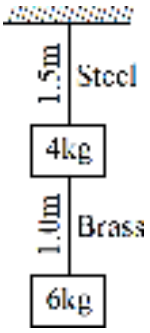
(1) 12 aT + mg

(2) aT – mg

(3) 12 aT

(4) 6aT + mg

45) Two wires of diameter 0.25 cm, one made of steel and other made of brass are loaded as shown in figure. The unloaded length of the steel wire is 1.5 m and that of brass is 1.0 m. Young's modulus of steel is 2.0×10^{11} Pa and that of brass is 1.0×10^{11} Pa. The ratio of elongations of steel and brass wires. $\left(\frac{\Delta \ell_{steel}}{\Delta \ell_{brass}} = ?\right)$



(1) 1.25

(2) 1.5

(3) 2.0

(4) 0.5

CHEMISTRY

1) The highest probability of finding the electron in an orbital having values of quantum numbers n = 3, □ = 1.

(1) In between any two axis out of three.

(2) On any two axis simultaneously, perpendicular to each other.

(3) On any one axis out of the three at a time.

(4) On all the three axis simultaneously at a time.

2) The set of ions which has same magnetic moment?

(1) Co^{+2}, Cr^{+3}, V^{+3}

(2) Mn^{+2}, Fe^{+3}, Cr^{+}

(3) Ni^{+2}, Mn^{+2}, Co^{+2}

(4) Fe^{+2}, Mn^{+2}, Co^{+2}

3) Ratio of wavelength of photon corresponding to γ-line of Paschen series in Li^{2+} ion and β line of Balmer series in He^{+} is -

(1) 1 : 1

(2) 1 : 2

(3) 1 : 4

(4) 3 : 16

4) What will be the ratio of de-Broglie wavelengths for two electrons which are accelerated through 100 volts and 400 volts respectively ?

(1) 2 : 1

(2) 1 : 4

(3) 1 : 2

(4) 4 : 1

5) The radius of any two of the first four Bohr's orbits of the hydrogen atom are in the ratio 1 : 4. The energy difference between them may be :-

(1) Either 12.09 eV or 3.4 eV

(2) Either 2.55 eV or 10.2 eV

(3) Either 13.6 eV or 3.4 eV

(4) Either 3.4 eV or 0.85 eV

6) In a hydrogen like species , an electron jumps from an orbital having two radial and two angular nodes to the orbital having same sign and value of wave function in all direction, at any particular distance. If the energy of emitted photon is 326.4 eV, the atom or ion is

(1) H

(2) Li^{2+}

(3) He^{+}

(4) B^{4+}

7)

Atoms $_{7}X^{A}$,$_{8}Y^{B}$,$_{9}Z^{17}$ are such that $_{8}Y$ is an isobar of $_{7}X$ and $_{9}Z^{17}$ atom is isotone of $_{8}Y$. Mass number of 'X' & number of neutrons in Y are respectively

(1) 8, 8

(2) 17, 7

(3) 9, 8

(4) 16, 8

8) The number of photons in a radiation of wavelength 662.6 nm that provide 3 J of energy are :

(1) 10^{9}

(2) 10^{19}

(3) 10^{28}

(4) 0.33×10^{9}

9) α-particles are projected towards the following metals, with the same kinetic energy. Towards which metal, the distance of closest approach is minimum.

(1) Cr

(2) Ag

(3) Cu

(4) Pt

10) The uncertainty in position of an electron is equal to its de-broglie wavelength. The minimum percentage error in its measurement of velocity under this circumtance will be approximately:-

(1) 4

(2) 8

(3) 22

(4) 18

11) Which of the following does not show conductivity ?

(1) Na(s)

(2) $AlF_3(\square)$

(3) $KClO_4$(aq.)

(4) $HCl(\square)$

12) In which of the following conversion both hybridisation and shape changes with respect to central atom?

(1) $H_2O \rightarrow H_3O^+$

(2) $2ClO_3 \rightarrow Cl_2O_6$

(3) $2NO_2 \rightarrow N_2O_4$

(4) $2CH_3 \rightarrow C_2H_6$

13) Correct order of decreasing adjacent bond angle is :-

(1) $XeO_3 > BrF_5 > XeF_4$

(2) $XeO_3 > XeF_4 > BrF_5$

(3) $XeO_3 < BrF_5 > XeF_4$

(4) $XeF_4 > XeO_3 > BrF_5$

14) Which of the following shows correct order of ionic character ?

(1) ZnO < ZnS

(2) $PbCl_2 < PbCl_4$

(3) AgCl < AgBr

(4) $PCl_3 > PCl_5$

15) Identify the iso-structural pair from the following

(i) $SiCl_4$, $SnCl_4$ (ii) NH_3, NO_3^-

(iii) NH_3, $ClO_3^{\ominus}$ (iv) CO_2, SO_2

(1) (i), (ii)

(2) (ii), (iii)

(3) (ii), (iv)

(4) (i), (iii)

16) Which of the following order is correct ?

(1) CO > CO_2 (C–O Bond length)

(2)  (Bond strength)

(3) $ClO_4^- > SO_4^{-2} > PO_4^{-3}$ (Bond order)

(4) LiF > Li_2O (Lattice energy)

17) Identify the incorrect statement in the following.

(1) The colour intensity of PbI_2 is greater than $PbCl_2$

(2) The acidic character of Mn_2O_7 is greater than MnO_2

(3) $HgCl_2$ having greater covalent character than $ZnCl_2$

(4) Melting point of KCl is more than KF.

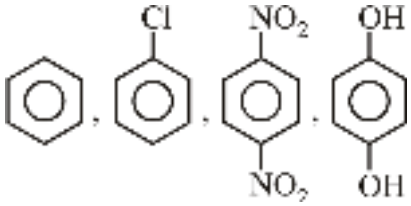
18) **Assertion (A) :-** $SeCl_4$ does not have a regular tetrahedral structure.
Reason (R) :- Se in $SeCl_4$ has two lone pairs.

(1) Both (A) & (R) are true & (R) is the correct explanation of (A)

(2) Both (A) & (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) & (R) are false

19) How many of the following are non-polar ?  , XeF_4, PCl_3F_2, XeF_6, N_2O_4

(1) 3

(2) 4

(3) 5

(4) 6

20) **Assertion :-** Boiling point of H_2SO_4 is more than HNO_3.
Reason :- H_2SO_4 have more extent of intra molecular H-bonding than HNO_3.

(1) Both assertion and reason are true but reason is not correct explanation of assertion.

(2) Assertion is true but reason is false.

(3) Assertion is false but reason is true.

(4) Both assertion and reason are true but reason is a correct explanation of assertion.

21) Consider the following statements & select the correct option.
Statement I : A polyatomic molecule can have zero dipole moment.
Statement II : It is necessary for a polyatomic molecule to have linear geometry to exhibit dipole

moment zero.

(1) Statement I is correct & statement II is incorrect.
(2) Statement II is correct & statement I is incorrect.
(3) Both the statements are correct.
(4) Both the statements are incorrect.

22) Which of the following is correct ?

	Compound	Types of bond length
(1)	H_2O	4
(2)	$Cr_2O_7^{2-}$	3
(3)	SF_6	2
(4)	PCl_5	2

(1) 1
(2) 2
(3) 3
(4) 4

23) **Assertion :** Ionic compounds exhibits electrical conductivity in solution state.
Reason : In solution state electrons of ionic compounds are free.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

24) $XeOF_4$ contain which type of π-bond?

(1) p_π–p_π
(2) p_π–d_π
(3) d_π–d_π
(4) both 2 and 3

25) **Assertion (A) :** F_2 is gas while I_2 is solid at room temperature.
Reason (R) : A larger molecule or heavy atom is more polarizable and has larger dispersion force (vander wall forces)

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

26) In ICl_2^+ , ICl_2^- and ICl_4^- sum of bond pair and lone pair on each iodine atom are respectively -

(1) 2, 2 and 4

(2) 2, 3 and 2

(3) 4, 5 and 4

(4) 4, 5 and 6

27) The geometry and hybridisation of SCl_4 will be

(1) T – shape, sp^3

(2) Distorted tetrahedral, sp^3d

(3) Tetrahedral, sp^3

(4) Trigonal bipyramidal, sp^3d

28)

List-I (Pair of species)		**List-II (Same characteristics in pair)**	
(P)	(CH_4,NH_3)	(1)	d-orbital participation in hybridisation of central atom is observed
(Q)	(PF_5, $IC\square_3$)	(2)	Tetrahedral electronic geometry of central atom
(R)	($BeC\square_2$,SO_3)	(3)	Atleast one pπ–pπ bond is present
(S)	(CO_2, SO_2)	(4)	Planar
		(5)	All possible bond angles are ≥ 90°

The correct code is :-

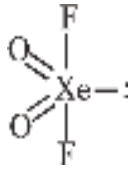
(1) P→2,4; Q→2; R→4,5; S→2,3,5

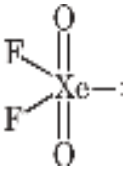
(2) P→2,3; Q→3; R→1,3; S→1,4,5

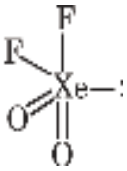
(3) P→2,5; Q→1; R→4,5; S→3,4,5

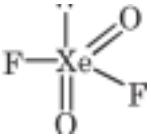
(4) P→2,5; Q→1; R→1,2; S→2,4,5

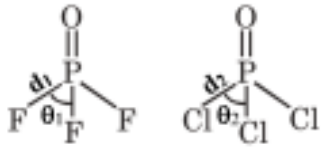
29) According to VSEPR method which of the following structure of XeO_2F_2 is **CORRECT**.

(1) 

(2) 

(3) 

(4) 

30) 

(θ_1, θ_2 = bond angle)
d_1, d_2 are P-F and P-Cl bond length respectively.
Choose correct order of θ_1, θ_2 and d_1, d_2 both

(1) $\theta_1 > \theta_2$, $d_2 > d_1$

(2) $\theta_2 > \theta_1$, $d_2 > d_1$

(3) $\theta_1 < \theta_2$, $d_1 > d_2$

(4) $\theta_1 > \theta_2$, $d_1 > d_2$

31) Which pair of element have same number of electron in their valence shell :-

(1) Na, Sr

(2) Se, Te

(3) Mn, Fe

(4) He, Ne

32) Atomic number of element E is 56. What should be the formula of its halide ?

(1) EX

(2) E_2X

(3) EX_2

(4) E_2X_3

33) Correct order of IP :-

(1) Zn > Ga

(2) Cd > In

(3) Hg > Tl

(4) All

34) Incorrect statement for the given configuration :- $[Xe]4f^{7}5d^{1}6s^{2}$

(1) It is placed in d-block, IIIB group, 6th period

(2) It is a natural element

(3) It has maximum unpaired e^{-} in lanthanide series.

(4) This element is Gd

35) Match the column

Column-I		**Column-II**	
(Atomic number)		**(Position of element in Periodic table)**	
(A)	Z = 37	(P)	p-block
(B)	Z = 42	(Q)	f-block
(C)	Z = 34	(R)	d-block
(D)	Z = 92	(S)	s-block

(1) A-P, B-Q, C-S, D-R

(2) A-P, B-Q, C-R, D-S

(3) A-S, B-R, C-P, D-Q

(4) A-S, B-R, C-Q, D-P

36) Which of the following is correct size order?

(1) N < C < Si < P

(2) Be < B < Al < Mg

(3) Cu < Zn < Ba < K

(4) F < O < S < Cl

37) Which of the following electron affinity order is incorrect ?

(1) Li > Na > K > Rb > Cs

(2) O > S > Se > Te

(3) Cl > F > Br > I

(4) Li > B > Be

38) Which of the following order of ionic radius is correct?

(1) $La^{+3} < Pm^{+3} < Ce^{+3} < Yb^{+3}$

(2) $La^{+3} < Ce^{+3} < Pm^{+3} < Yb^{+3}$

(3) $Yb^{+3} < Ce^{+3} < Pm^{+3} < La^{+3}$

(4) $Yb^{+3} < Pm^{+3} < Ce^{+3} < La^{+3}$

39) Incorrect set is :-
(a) IP (Na < Li < B < Be)

(b) EA (N < P < O < S)
(c) EN (S < O < Cl < F)
(d) Size (F < N < O < Na)

(1) a
(2) a, b
(3) a, b, c
(4) c, d

40) What is the correct order of ionisation energy ?

(1) B > Al < Ga > In < Tl
(2) B < Al > Ga > In > Tl
(3) B > Al > Ga > In > Tl
(4) B > Al ≤ Ga < In < Tl

41) Which is correct order ?

(1) HF < HCl < HBr < HI (Acidic nature)
(2) HOCl > HOBr > HOI (Acidic nature)
(3) NaOH > $Mg(OH)_2$ (Basic nature)
(4) All of these

42) The correct order of electronegativity of N, P, C and Si is ?

(1) N < P < C < Si
(2) P > N > Si > C
(3) N = P > C = Si
(4) N > C > P > Si

43) Correct order of EA is :

(1) O > Cl > N > C
(2) Cl > C > O > N
(3) Cl > O > C > N
(4) N > C > O > Cl

44) Given below are two statements:
Statement I : Fluorine has most negative electron gain enthalpy in its group.
Statement II : Oxygen has least negative electron gain enthalpy in its group.
In the light of the above statements, choose the most appropriate from the options given below

(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is false but Statement II is true
(4) Statement I is true but Statement II is false

45) **Assertion :** Electronegativity of nitrogen is greater than carbon.
Reason : Nitrogen has stable half filled electronic configuration.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BIOLOGY

1) Correctly match column-I, II and III.

	Column-I		**Column-II**		**Column-III**
(A)	Golgi apparatus	(P)	Chromatin	(i)	Protein synthesis
(B)	Ribosome	(Q)	Cristae	(ii)	Packaging and secretion
(C)	Nucleus	(R)	Cisternae	(iii)	Controller of cell
(D)	Mitochondria	(S)	RNA and proteins	(iv)	Aerobic respiration

(1) (A)-(P)-(i), (B)-(Q)-(ii), (C)-(R)-(iii), (D)-(S)-(iv)

(2) (A)-(R)-(ii), (B)-(P)-(i), (C)-(S)-(iii), (D)-(Q)-(iv)

(3) (A)-(R)-(ii), (B)-(S)-(i), (C)-(P)-(iii), (D)-(Q)-(iv)

(4) (A)-(S)-(ii), (B)-(R)-(i), (C)-(P)-(iii), (D)-(Q)-(iv)

2) Read the following statements :
(a) The mitochondria divide by fission
(b) The ribosomes of the chloroplast are smaller (70S) than the cytoplasmic ribosomes (80S)
(c) Ribosomes are composed of ribonucleic acid (RNA) and proteins and are not surrounded by any membrane
(d) Nucleus as a cell organelle was first described by Robert Brown.
(e) Acrocentric chromosome has a terminal centromere.
Choose the correct statements :

(1) a, b, d & e

(2) b, c, d & e

(3) a, c, d & e

(4) a, b, c & d

3) Match the column-I and column-II and select the correct option :–

	Column-I		**Column-II**
(A)	Hydrolytic enzymes	(i)	Gives the appearance of small fragment
(B)	Osmoregulation & excretion	(ii)	Present in prokaryotic cell
(C)	Peripheral triplet	(iii)	Centriole
(D)	Non staining secondary constriction	(iv)	Centromere

		(v)	pH 5
		(vi)	Vacuole
		(vii)	Basal body

(1) (A)-(v), (B)-(vi), (C)-(iii),(vii), (D)-(i)

(2) (A)-(v),(vi), (B)-(iii), (C)-(i),(iv), (D)-(ii),(vii)

(3) (A)-(ii),(v), (B)-(vi), (C)-(iii),(vii), (D)-(i),(iv)

(4) (A)-(v), (B)-(vi),(vii), (C)-(iv), (D)-(i),(iii)

4) Given below are two statements :
Statement-I :- The cytoskeleton in a cell are involved in mechanical support, DNA replication, motility etc.
Statement-II :- An elaborate network of filamentous proteinaceous structures consisting of microtubules, microfilaments and intermediate filament present in the cytoplasm is collectively referred to as the cytoskeleton.
In the light of above statements, choose the correct answer from the options given below :

(1) Statement I is incorrect but statement II is correct

(2) Statement I is correct but statement 2 is incorrect

(3) Both the statements are incorrect

(4) Both the statements are correct

5) Match the given organelle in column-I with their related functions in column-II and select correct option -

	Column-I		**Column-II**
(A)	Smooth endoplasmic reticulum	(I)	Lipid synthesis
(B)	Lysosome	(II)	Hydrolysis of food
(C)	Vacuole	(III)	Storage of secretary & waste product
(D)	Rough endoplasmic reticulum	(IV)	Actively involved in protein synthesis and secretion

(1) (A)→I ; (B)→II ; (C)→III ; (D)→IV

(2) (A)→III ; (B)→IV ; (C)→I ; (D)→II

(3) (A)→IV ; (B)→III ; (C)→II ; (D)→I

(4) (A)→II ; (B)→I ; (C)→III ; (D)→IV

6) **Statement-I :** Schwann concluded based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells.
Statement-II : Cells that have membrane bound nuclei are called eukaryotic where as cells that lack membrane bound nucleus are considered prokaryotic.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct Statement II is incorrect.

(4) Statement I is incorrect Statement II is correct.

7) Select the incorrect statement :-

(1) Centromere holds two chromatids of a chromosome.

(2) The shorter and longer arms of a submetacentric chromosomes are referred to as p-arm and q-arm respectively.

(3) A single human cell has approximately two meter long thread of DNA.

(4) In acrocentric chromosome the centromere is slightly away from the middle of the chromosome.

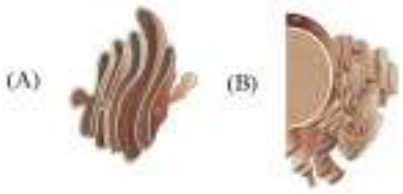
8) Membrane bound vesicular structures which contain digestive acid hydrolases are formed by :-

(1) Lysosome

(2) Golgi body

(3) Microbodies

(4) Mitochondria



9) Identify the figures gives below A & B and choose correct answer :

(1) (A) ER→Lipid synthesis, (B) GB→ Formation of glycoprotein

(2) (A) ER→Formation of glycoprotein, (B) GB→ Lipid synthesis

(3) (A) GB→ Formation of glycoprotein, (B) ER→ Lipid synthesis

(4) (A) GB→ Steroidal hormone synthesis, (B) ER→ Power house of the cell

10) The fluid nature of the membrane is important from the point of view of functions like -

(1) Cell division & cell growth

(2) Endocytosis & secretion

(3) Formation of intercellular junctions

(4) All of the above

11) Which of the following enzyme is not present in lysosomes?

(1) Proteases

(2) Lipases

(3) Nucleases

(4) Catalase

12) Reductionist Biology is:

(1) The concepts and techniques of physics and chemistry are applied to understand biology.

(2) Only physical approach to study and understand living organisms.

(3) Physiological and behavioural approach to study and understand living organisms.

(4) Only chemical approach to study and understand living organism.

13) The stroma of chloroplast does not contain ?

(1) Enzymes for the synthesis of carbohydrates and proteins

(2) Small, double stranded, linear DNA

(3) Ribosomes

(4) ATP and NADPH + H^+

14) Choose the correct statement :-

(1) Chromatin name was given by flemming.

(2) Nucleus is absent in mature sieve tube and WBC.

(3) Nuclear lamina structure is formed by filament of tubulin protein.

(4) The nuclear pore is guarded by two octagonal discoid structure of actin protein.

15) Which of the following statement is wrong with respect to mitochondria ?

(1) Mitochondria surrounded by two membrane outer and inner membrane.

(2) Mitochondria is also called power houses of the cell.

(3) The inner compartment of mitochondria is filled with a dense homogenous substance called the matrix.

(4) The matrix possesses circular DNA, few RNA and 80 s type of ribosomes.

16) **Assertion (A) :** Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S-phase.
Reason (R) : Meiosis involves pairing of homologous chromosomes and recombination between homologous chromosomes.

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true and R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

17) Which of the following statements is not correct with respect to anaphase?

(1) At the onset of anaphase, each chromosome arranged at the metaphase plate is split simultaneously.

(2) The two daughter chromatids, now referred to as daughter chromosomes of the future daughter nuclei, begin their migration towards the two opposite poles.

(3) As each chromosome moves away from the equatorial plate, the centromere of each chromosome remains directed towards the pole and hence at the leading edge, with the arms of the chromosome trailing behind.

(4) During anaphase, centromeres do not split and chromatids do not separate.

18) A stage in a cell division give in a figure :  Select the correct answer :

(1) Telophase :- two cells have two daughter nuclei
(2) Metaphase :- two cells have two daughter nuclei
(3) Telophase :- one cell have two daughter nuclei
(4) Anaphase :- one cell have two daughter nuclei

19) Match the following :-

Column-I		**Column-II**	
(a)	Prophase	(i)	Chromosomes moved to spindle equator
(b)	Anaphase	(ii)	Chromosomal material condensation start
(c)	Metaphase	(iii)	Chromosome material decondense
(d)	Telophase	(iv)	Centromere splits and chromatids separate

(1) a - (ii), b - (iv), c - (i), d - (iii)
(2) a - (iii), b - (iv), c - (i), d - (ii)
(3) a - (iv), b - (iii), c - (ii), d - (i)
(4) a - (ii), b - (i), c - (iii), d - (iv)

20) If a cell has 24 chromatids after s-phase, what will be the number of chromosome at each pole in Anaphase-I ?

(1) 6
(2) 12
(3) 24
(4) 3

21) Cells remain metabollically active but no longer proliferate (divide) unless called on to do so depending on the requirement of the organism.
This is correct for :-

(1) G_1 phase
(2) G_2 phase
(3) S-phase
(4) G_0 phase

22) Which of following event does not take place during S phase of cell cycle ?

(1) DNA replication

(2) Centriole duplication

(3) Cell growth

(4) Tubulin protein synthesis

23) The anaphase promoting complex involved in the separation of ___(i)___ and, ___(ii)___ of chromosomes and ___(iii)___ of nuclear membrane.

(1) (i) Sister chromatids, (ii) Decondensation, (iii) Reassembly

(2) (i) Non-sister chromatids, (ii) Decondensation, (iii) Reassembly

(3) (i) Sister chromatids, (ii) Condensation, (iii) Disassembly

(4) (i) Non-sister chromatids, (ii) Condensation, (iii) Reassembly

24) Which one of the following stage of meiosis begins with the spliting of centromere?

(1) Anaphase

(2) Metaphase II

(3) Anaphase II

(4) Telophase II

25) Mitosis accomplishes not only the segregation of duplicated chromosomes into daughter nuclei but the cell itself is divided into two daughter cells by a separate process called :-

(1) Karyokinesis

(2) Cytokinesis

(3) Syncytium

(4) Fertilisation

26) In an onion root tip cell number of chromosomes and DNA content at G_1 stage are 16 and 2C respectively. After s-phase number of chromosomes and DNA content respectively will be :

(1) 16, 2C

(2) 16, 4C

(3) 4C, 16

(4) 32, 4C

27) **Statement-I :-** Meiosis is the mechanism by which conservation of specific chromosome number of each species is achieved across generation in sexually reproducing organisms, even through the process (per se paradoxically) results in reduction of chromosome number by half.
Statement-II :- Meiosis also increase the genetic variability in the population from one generation to the next generation and genetic variations are very important for the process of evolution.

(1) Both **Statement-I** and **Statement-II** are incorrect.

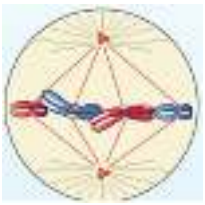
(2) **Statement-I** is correct but **Statement-II** is incorrect.

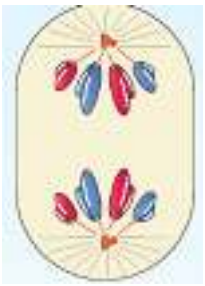
(3) **Statement-I** is incorrect but **Statement-II** is correct.

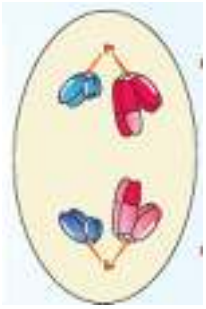
(4) Both **Statement-I** and **Statement-II** are correct.

28) Which of the following diagram is not associated to mitosis ?

(1) 

(2) 

(3) 

(4) 

29) Few events regarding cell cycle are given below. Arrange these events in correct order and select the correct option from option given below:
A. Chromosomes replication
B. Centromere division
C. Cell organelle synthesis
D. Condensation of chromatin material complete
E. Division of cytoplasm

(1) C,B,A,D,E

(2) C,D,B,A,E

(3) C,A,D,B,E

(4) C,D,A,B,E

30) Fill up the blanks in the following statements with the appropriate options.
(I) The complex formed by a pair of synapsed homologous chromosomes is called as _A_ .
(II) Enzyme involved in crossing over is _B_ .
(III) The X-shaped structure formed during diplotene is _C_ .
(IV) Dyad of cells are formed at the end of _D_ .

	A	B	C	D

(1)	Synapsis	Permease	Synaptonemal complex	Telophase-I
(2)	Bivalent	Recombinase	Chiasmata	Meiosis-I
(3)	Tetrad	Lyase	Bivalent	Meiosis-I
(4)	Nodules	Endonuclease	Chiasmata	Anaphase-I

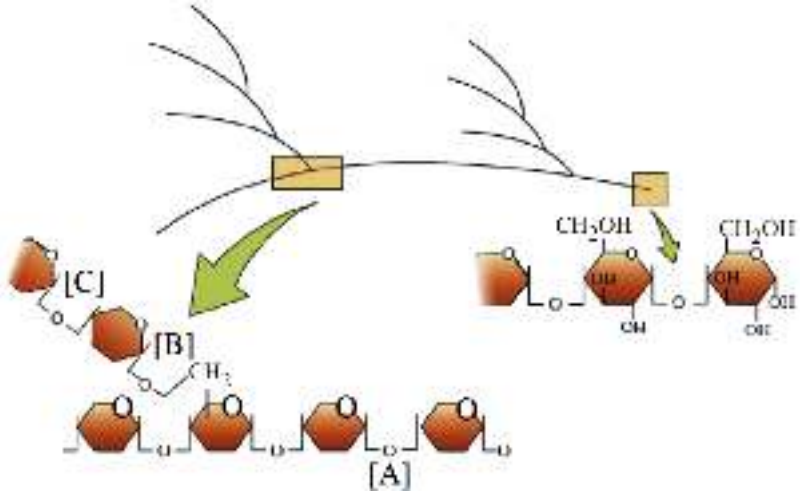
(1) 1

(2) 2

(3) 3

(4) 4

31) In the given below diagrammatic representation of a portion of glycogen, glycosidic bonds



represented by A, B and C are :-

	A	B	C
(1)	α-1'-4"	α-1'-4"	β-1'-4"
(2)	β-1'-6"	β-1'-4"	β-1'-4"
(3)	α-1'-6"	α-1'-4"	α-1'-4"
(4)	α-1'-4"	α-1'-6"	α-1'-4"

(1) 1

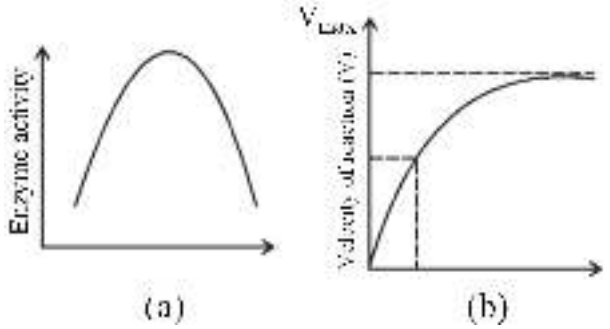
(2) 2

(3) 3

(4) 4

32)

Consider following graphs regarding the effect of change of different factors on enzyme activity
Identify the factor (a) and (b)



(1) (a) - Substrate concentration, (b) - temperature

(2) (a) - pH, (b) - substrate concentration

(3) (a) - Temperature, (b) - pH

(4) (a) - Substrate concentration, (b) - pH

33) Match the column-A to column-B

Column-A		**Column-B**	
(A)	Apoenzyme	(I)	Tightly bond
(B)	Prosthetic group	(II)	Co-enzyme
(C)	Co-factor	(III)	Hydrogen peroxide
(D)	Catalase	(IV)	Protein part

(1) A - II, B - I, C - IV, D - III

(2) A - IV, B - I, C - II, D - III

(3) A - IV, B - I, C - III, D - II

(4) A - I, B - II, C - III, D - IV

34) Given below are two statements :
Statement-I : Low temperature keep the enzyme in a permanently inactive state whereas high temperature destroys enzymatic activity because proteins are denatured by heat and cold.
Statement-II : When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is called as noncompetitive inhibitor.
In the light of above statements, choose the correct answer from the options given below :

(1) Both **Statement I** and **Statement II** are false

(2) **Statement I** is true and **Statement II** is false

(3) **Statement I** is false and **Statement II** is true

(4) Both **Statement I** and **Statement II** are true

35) Match the list-I with list-II :

	List-I		**List-II**
(A)	Acidic amino acid	(I)	Valine
(B)	Basic amino acid	(II)	Glutamic acid
(C)	Neutral amino acid	(III)	Phenyl alanine

(D)	Aromatic amino acid	(IV)	Lysine

Choose the correct answer from the options given below :

(1) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(2) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(3) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(4) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

36) Read the following statements and choose the incorrect statements.
(a) Palmitic acid has 16 carbons excluding carboxyl carbon.
(b) Lecithin is a phospholipid.
(c) Lipids are not strictly macromolecules.
(d) Cellulose is a heteropolymer.
(e) Protein is a homopolymer.
Choose the correct options :

(1) (a), (b) and (c)

(2) (c), (d) and (e)

(3) (b), (c) and (d)

(4) (a), (d) and (e)

37) Given below are two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A) :- In a polysaccharide chain, the right end is called the reducing end and the left end is called the non reducing end.
Reason (R) :- Polysaccharides are usually non reducing carbohydrates.
In the light of above statements, choose the correct answer from the options given below :-

(1) A and R both are true but R is not correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) A and R both are true and R is the correct explanation of A.

38) How many total number of carbon present in Arachidonic acid?

(1) 14

(2) 16

(3) 18

(4) 20

39) Which structure of protein show hollow woolen ball like structure?

(1) Primary Structure

(2) Secondary Structure

(3) Tertiary Structure

(4) Quaternary Structure

40) Which of the following is not an example of macro-molecules ?

(1) Polysaccharides

(2) Proteins

(3) Nucleic acids

(4) Lipids

41) Select the correct match-

	A	B
(1)	Trypsin	Hormone
(2)	Collagen	Enzyme
(3)	Insulin	Sensory reception
(4)	Antibody	Fights Infectious Agents

(1) 1

(2) 2

(3) 3

(4) 4

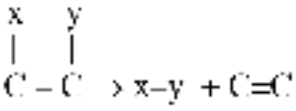
42) Starch forms blue colour with iodine because :-

(1) It is a linear molecule.

(2) It contains complex helices and hence hold iodine molecules.

(3) It breaks down when iodine reacts with it.

(4) It is a disaccharide.

43) 

the above reaction catalysed by which class of enzyme:-

(1) Ligases

(2) Transferases

(3) Lyases

(4) Hydrolases

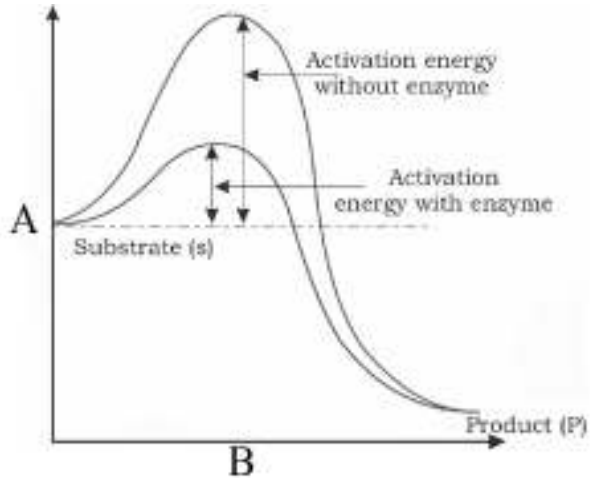
44) The enzymes that catalyse hydrolysis of ester, ether, peptide, glycosidic, C—C, C-halide or P-N bonds belong to class ?

(1) Hydrolases

(2) Lyases

(3) Isomerases

(4) Ligases



45)

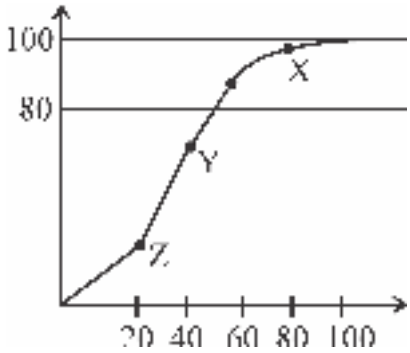
A in the above graph represents-

(1) Progression of reaction

(2) Potential energy

(3) Transition state

(4) Activation energy



46) Which one is correct statement for oxygen dissociation curve ?

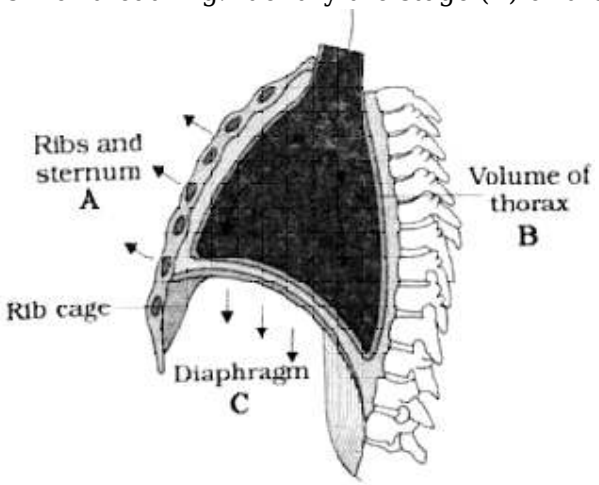
(1) X → dissociation of oxygen at alveoli level

(2) Y → Intensive exercise condition for tissue

(3) Y → 100 ml at blood gives aprox 15 ml O_2 to tissue

(4) Z → 100 ml at blood gives 15 ml O_2 to tissue

47) The figure given below shows the mechanism of breathing. Identify the stage (X) of breathing



explained & A, B and C marked in the figure.

(1) X – Expiration, A – raised, B – decreased, C – relaxed

(2) X – Inspiration, A – raised, B – decreased, C – relaxed

(3) X – Expiration, A – raised, B – increased, C – contracted

(4) X – Inspiration, A – raised, B – increased, C – contracted

48) Match the columns and choose the correct options :

	Column-I		**Column-II**		**Column-III**
(A)	Granulocyte	(P)	Neutrophil	(i)	Heparin secretion
(B)	Agranulocyte	(Q)	Lymphocyte	(ii)	Allergic Reaction
		(R)	Basophil	(iii)	Phagocytosis
		(S)	Acidophil	(iv)	Immune response

	i	ii	iii	iv
1	R,A	S,B	P,A	Q,B
2	R,A	S,A	P,A	Q,B
3	R,B	S,B	P,A	Q,A
4	R,A	S,B	P,B	Q,B

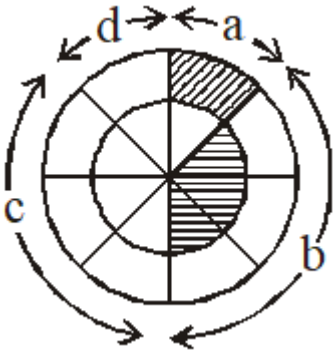
(1) 1

(2) 2

(3) 3

(4) 4

49) Which of the following statement is correct about the given diagram of cardiac cycle ?



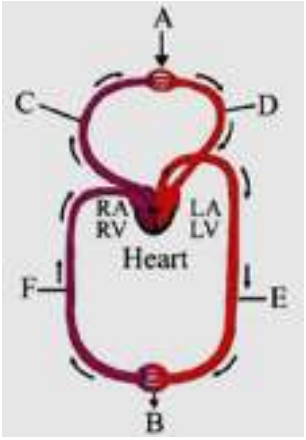
(Note :- Shaded part indicate systole)

(1) 'a' represent atrial systole during which 25% blood moves from atria to ventricles actively and slowly.

(2) (a → b → c) represents (Atrial systole → ventricular systole → joint diastole)

(3) 'c' and 'd' together called as joint diastole during which approx 75% of blood moves from atria to ventricles passivelly.

(4) 'b' represents phase of diastasis during which rate of filling of ventricles is twice to the rate of flow of blood into atria.

50) Identify A to F in given diagram :- 

Choose the correct option.

(1) A-Lungs, B-Body parts, C-Pulmonary vein, D-Pulmonary artery, E-Dorsal aorta, F-Vena cava

(2) A-Lungs, B-Body parts, C-Pulmonary artery, D-Pulmonary vein, E-Dorsal aorta, F-Vena cava

(3) A-Lungs, B-Body parts, C-Pulmonary artery, D-Pulmonary vein, E-Vena cava, F-Dorsal aorta

(4) A-Body parts, B-Lungs, C-Pulmonary artery, D-Pulmonary vein, E-Vena cava, F-Dorsal aorta

51) **Assertion :** A special case of Rh incompatibility (mismatching) in erythroblastosis foetalis has been observed between the Rh– blood of a pregnant mother with Rh+ blood of the foetus.
Reason : In case of subsequent pregnancies, the Rh antibodies from the mother (Rh–) can leak into the blood of the foetus (Rh+) and destroy the foetal RBCs.

(1) Assertion and reason both are true and the reason is correct explanation of assertion.

(2) Assertion and reason both are true and the reason is not correct explanation of assertion.

(3) Assertion is true but reason is wrong.

(4) Assertion and reason both are wrong.

52) In the summer season, a person is walking on the road in the afternoon, causing excessive sweating. Which of the following mechanisms helps maintain the osmolarity of body fluids despite the loss of water due to sweating?

(1) Activation of osmoreceptors which can stimulate the hypothalamus to release antidiuretic hormone from the neurohypophysis.

(2) Activation of osmoreceptors which can stimulate the hypothalamus to release antidiuretic hormone from the adenohypophysis

(3) Activation of osmoreceptors which can stimulate the hypothalamus to release atrial natriuretic factor from the neurohypophysis

(4) Activation of osmoreceptors which can stimulate the JG cells to release antidiuretic hormone from the neurohypophysis

53) Which one of the following options gives the correct categorisation of six animals according to the type of nitrogenous wastes (A, B, C), they give out ?

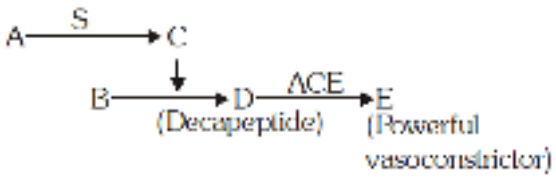
	A AMMONOTELIC	B UREOTELIC	C URICOTELIC
(1)	Aquatic Amphibia	Frog, Humans	Pigeon, Lizards, Cockroach
(2)	Aquatic Amphibia	Cockroach, Humans	Frog, Pigeon, Lizards
(3)	Pigeon, Humans	Aquatic Amphibia, Lizards	Cockroach, Frog
(4)	Frog, Lizards	Aquatic Amphibia, Humans	Cockroach, Pigeon

(1) 1

(2) 2

(3) 3

(4) 4



54) Go through the following diagram– If S = A fall is GFR or glomerular b.p.

A = J G cells

Identify the organ which secretes 'B'–

(1) Kidney

(2) Liver

(3) Lung

(4) Adrenal gland

55) Match the columns and choose the correct option

Column-I		Column-II		Column-III	
P	Cartilagenous box	A	irregular walled	I	Larynx
Q	Vascularised bag	B	Cartilagenous	II	Alveoli
R	Flap like	C	Sound box	III	Epiglottis

(1) P-C-I, Q-A-II, R-B-III

(2) P-A-III, Q-C-II, R-B-I

(3) P-A-II, Q-B-III, R-C-I

(4) P-B-I, Q-A-II, R-C-III

56) Match the following correctly.

	Animals		**Respiratory organ**
(A)	Earthworm	[I]	Lungs
(B)	Mollusca	[II]	trachea
(C)	Fishes	[III]	Gills
(D)	Birds/Reptiles	[IV]	Moist cuticle
(E)	Insects		

(1) A-IV, B and C-III, D-I, E-II

(2) A-IV, B-III, C and D-I,, E-III

(3) A-II, B and C-III, D-I, E-IV

(4) A-III, B and C-I, D-II, E-IV

57) **Statement-I :** During respiration, O_2 is utilized by the organism to directly break down nutrient molecules.
Statement-II : The process of exchange of O_2 from the atmosphere with CO_2 produced by the cells is commonly known as respiration.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

58) Which of the following statement is correct about mechanism of breathing ?

(1) Inspiration is initiated by relaxation of diaphragm

(2) Contraction of diaphragm increases the volume of thoracic chamber in antero-posterior axis.

(3) IICM and diaphragm are the chief muscle for inspiration.

(4) Normal expiration is an active process.

59) **Assertion :** Every 100 ml of oxygenated blood can deliver around 4 ml of O_2 to the tissues under normal physiological conditions.
Reason : Every 100 ml of deoxygenated blood delivers approximately 5 ml of CO_2 to the alveoli.

(1) Assertion and reason both are true and the reason is correct explanation of assertion.

(2) Assertion and reason both are true and the reason is not correct explanation of assertion.

(3) Assertion is true but reason is wrong.

(4) Assertion and reason both are wrong.

60) Read following statements carefully and choose the incorrect one.

(1) Binding of CO_2 with haemoglobin is related to the partial pressure of CO_2.

(2) pO_2 is major factor which could affect binding of CO_2 with haemoglobin.

(3) When pCO_2 is high and pO_2 is low as in the alveoli, dissocation of CO_2 from carbaminohaemoglobin take place.

(4) Approximately 4ml CO_2 delivers to the alveoli by every 100 ml of deoxygenated blood.

61) **Assertion (A) :** High blood pressure leads to heart disease and also affects vital organs like brain and kidney.
Reason (R) : On Repeated checks of BP of an individual is 140|90 or more represents the hypertension.

(1) (A) and (R) both are true and (R) is correct explanation of (A).

(2) (A) and (R) both are correct but (R) is not correct explanation of (A)

(3) (A) is true (R) is false

(4) (A) and (R) both are incorrect.

62) Match the column-I and column-II.

	Column-I		**Column-II**
a.	Joint diastole	(i)	0.8 second
b.	Heart beat	(ii)	All four chambers of heart are relaxed.
c.	Cardiac cycle	(iii)	72 times/min
d.	Stroke volume	(iv)	70 mL

Choose the correct answer about given column :-

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(ii), b-(iii), c-(i), d-(iv)

(3) a-(iii), b-(ii), c-(i), d-(iv)

(4) a-(iv), b-(iii), c-(ii), d-(i)

63) Given below are two statements :-
Statements-I : Plasma is a straw coloured, viscous fluid constituting nearly 90-92 percent of blood.
Statements-II : Reduction in platelets number can lead to clotting disorders.

(1) Statement-I is true but statement-II is false

(2) Statement-I is false but statement-II is true.

(3) Both statement-I and statement-II are true

(4) Both statement-I and statement-II are false.

64) Which of one of the following statement is correct about blood grouping ?

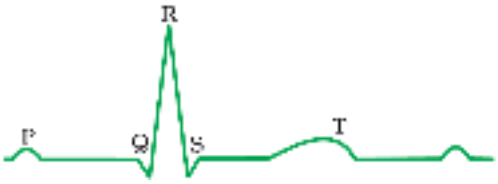
(1) ABO grouping is based on presence or absence of two surface antigens on RBCs namely A and O.

(2) Group 'AB' blood can be donated to persons with any other blood group.

(3) Mismatched blood transfusion cause severe problems of clumping.

(4) An Rh⊕ve person if exposed to Rh⊖ve blood, will form specific antibodies against the Rh antigens.

65) Which of the following option is correct regarding this figure :



(1) P wave represent repolarisation of atria

(2) Letter P to T shows chemical activity of heart

(3) Systole of auricle starts shortly after Q & marks the beginning of the contraction

(4) The T wave represents the return of ventricles from excited to normal state

66) Which of the following statement is incorrect?

(1) The renal tubule begins with a double walled cup-like structure called Bowman's capsule.

(2) In majority of nephrons, the loop of Henle is too short and extends only very little into the medulla.

(3) In juxta medullary nephrons, loop of Henle is very long and runs deep into the medulla.

(4) Towards the centre of inner convex surface of the kidney, a notch is present, through which ureter, blood vessels and nerves enter.

67) **Assertion (A) :** As a concentrated filterate pass upward through ascending limb of henle loop, it gets diluted.
Reason (R) : Ascending limb of henle loop transports electrolyte actively or passively to the medullary fluid.

(1) A is correct but R is not correct.

(2) A and R both are correct but R is not correct explanation of A.

(3) A and R both are correct and R is correct explanation of A.

(4) Both A & R are incorrect.

68) **Statement I :** Collecting duct of nephron extends from the cortex of the kidney to the inner parts of medulla.
Statement II : Large amount of water could be reabsorbed from collecting duct to form a concentrated urine.

(1) Statement I & II both are correct.

(2) Statement I & II both are incorrect.

(3) Statement I is correct but II is incorrect.

(4) Statement I is incorrect but II is correct.

69) Match column I with column II

Column I	**Column II**

(i)	Renal failure	(a)	inflammation of kidney.
(ii)	Diabets Mellitus	(b)	Stone or isoluble mass of crystalised salt
(iii)	Glomerulonephritis	(c)	Presence of glucose & ketone bodoes in urine
(iv)	Renal calculi	(d)	Malfunctioning of kidney

(1) i→a,ii→b,iii→c,iv→d

(2) i→c,ii→b,iii→d,iv→a

(3) i→d,ii→c,iii→a,iv→b

(4) i→a,ii→b,iii→c,iv→d

70) Match the Column-I with Column-II and select the correct option given below :

	Column-I (Function)		**Column-II (Part of excretory system)**
(a)	Counter current mechanism	(I)	Proximal convulated tubule
(b)	Maximum reabsorption	(II)	Distal convulated tubule
(c)	Urea reabsorption	(III)	Malphigian corpuscle
(d)	Ultrafiltration	(IV)	Henle's loop
		(V)	Collecting duct

	(a)	**(b)**	**(c)**	**(d)**
(1)	(IV)	(V)	(II)	(III)
(2)	(IV)	(I)	(V)	(III)
(3)	(V)	(IV)	(I)	(II)
(4)	(V)	(IV)	(I)	(III)

(1) 1

(2) 2

(3) 3

(4) 4

71) A chemosensitive area is highly sensitive to :-

(1) O_2 and H^+ ions

(2) CO_2 and Na^+ ions

(3) CO_2 and H^+ ions

(4) CO_2 and O_2

72) Most of the aquatic arthropods and molluscs respire through :-

(1) Network of tubes
(2) Gills
(3) Moist skin
(4) Lungs

73) In heavy smokers the alveoli of the lungs are enlarged and alveoli septa damaged which reduces the surface area of the exchange of respiratory gases. This condition is called :-

(1) Asthma
(2) Silicosis
(3) Emphysema
(4) Insomnia

74) If expiratory reserve volume is 1100 ml residual volume is 1200 ml and tidal volume is 500 ml, what shall be the functional residual capacity–

(1) 1600 ml
(2) 2800 ml
(3) 2300 ml
(4) 1200 m

75) Trachea divided at the level of _____ thoracic vertebra into primary bronchi

(1) 5th
(2) 6th
(3) 10th
(4) 4th

76) In a healthy adult man, the partial pressure of oxygen in the systemic arteries is 95 mm Hg which is :-

(1) Less than the pO_2 in lung alveoli
(2) Less than the pCO_2 in systemic veins
(3) Equal to the pCO_2 in systemic veins
(4) Less than the pO_2 in actively metabolising tissues

77) 'Inspiration occurs if pressure within the lungs is __A__ than the atmospheric pressure. Choose appropriate option to fill in the blank (A) in given statements?

(1) More
(2) Less
(3) Equal
(4) In inspiration air moves out of lungs

78) The average life span of a human RBC is __A__ and the organ considered graveyard of erythrocytes is __B__ choose the correct option that fills the blanks A and B respectively.

(1) 120 days, liver
(2) 180 days, liver
(3) 120 days, spleen
(4) 120 days, kidney

79) Which of following formed by cascade process to help in blood clotting.

(1) Thrombokinase
(2) Heparin
(3) Fibrinogen
(4) Prothrombin

80) In which of the following animals, the heart does not have the left and right auricles ?

(1) Mammals
(2) Reptiles
(3) Amphibians
(4) Fishes

81) Which of following nutrient absorbed through lymph in the lacteal present in intestinal villi -

(1) Fat
(2) Protein
(3) Carbohydrate
(4) All of above

82) Which layer of blood vessels is called endothelium ?

(1) Tunica intima
(2) Tunica media
(3) Tunica externa
(4) All of above

83) A unique Vascular Connection exists between the digestive tract and liver is called :-

(1) Digestive Portal System
(2) Renal Portal System
(3) Hepatic Portal System
(4) Coronary System

84) SA node is present in the :-

(1) right upper corner of the left atrium

(2) left upper corner of the right atrium
(3) right lower corner of the right atrium
(4) right upper corner of the right atrium

85) Glomerulus is a tuft of capillaries formed by the afferent arteriole which is a fine branch of

(1) Ventral aorta
(2) Renal vein
(3) Renal artery
(4) Inferior vena cava

86) Which of the following substance is not eliminated by secretion of sebaceous gland?

(1) Sterols
(2) Hydrocarbons
(3) Wax
(4) NaCl

87) Urination/micturition occurs when :

(1) Urinary bladder relaxes and urethral sphincter contracts
(2) Urinary bladder and urethral sphincter both contracts
(3) Urinary bladder contracts and urethral sphincter relaxes
(4) Urinary bladder and urethral sphincter both relaxes

88) The Hyperosmolarity gradient in medullary interstitium is mainly caused by :-

(1) NaCl only
(2) KCl
(3) Ammonia and NaCl
(4) Urea and NaCl

89) In which part of nephron minimum reabsorption occurs ?

(1) PCT
(2) Desending limb of LOH
(3) Ascending limb of LOH
(4) DCT

90) Auto-excitable heart of human :-

(1) Can't be moderated.
(2) Can be moderated by hyperventilation.
(3) Can be moderated by neural & hormonal mechanism.
(4) Can be moderated by external mechanical force.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	1	4	1	2	3	4	3	1	4	3	2	4	1	4	1	4	1	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	1	1	1	4	3	4	4	1	1	3	3	3	3	1	3	3	3	2
Q.	41	42	43	44	45															
A.	3	2	2	1	1															

CHEMISTRY

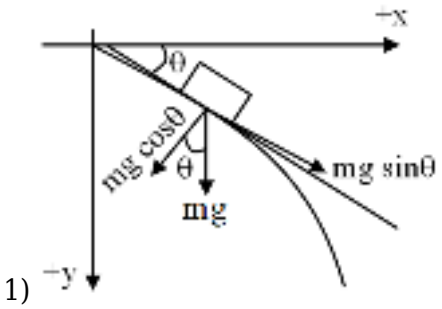
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	1	1	2	4	4	2	1	2	4	4	2	4	4	3	4	3	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	3	2	4	4	2	3	1	2	2	3	4	1	3	3	2	4	4	1
Q.	86	87	88	89	90															
A.	4	4	3	3	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	4	1	1	1	1	4	2	3	4	4	1	2	1	4	2	4	3	1	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	4	1	3	2	2	4	4	3	2	4	2	2	1	1	4	1	4	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	3	1	2	4	4	2	3	2	1	1	1	2	1	1	4	2	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	2	3	4	4	3	1	3	2	3	2	3	3	1	1	2	3	1	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	1	3	4	3	4	3	4	3	3										

SOLUTIONS

PHYSICS



1)

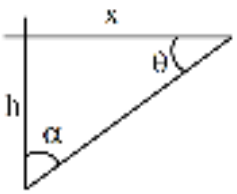
Here θ is + ve because lower side is +ve axis

$y = \frac{x^2}{2}$

$\frac{dy}{dx} = x = \tan\theta$

tangential acceleration $a_t = g\sin\theta = \frac{gx}{\sqrt{1+x^2}}$

2) $\omega = \frac{dx}{dt} = 0.3$ rad/sec



$\tan\alpha = \frac{x}{h} \Rightarrow \sec^2\alpha \frac{d\alpha}{dt} = \frac{1}{h}\frac{dx}{dt}$

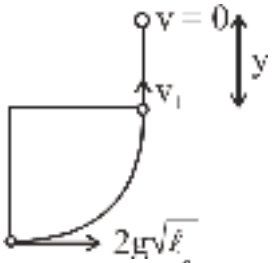
$(\sec^2\alpha)(\omega) = \frac{1}{h}(v_p)$

$(\sec^2 45)(0.1) = \frac{1}{3}(v_p)$

$v_p = 0.6$ rad/sec.

3) By energy conservation

$v_1 = \sqrt{2g\ell}$ and $y = \frac{(2g\ell)}{2g} = \ell$



distance $= \frac{\pi\ell}{2} + \ell = \frac{14}{9}\left[\frac{11}{7} + 1\right] = 4m$

4)

$\theta = \omega_i t + \frac{1}{2}\alpha t^2$

$\theta = 0 + \frac{1}{2}(4)(4)^2$

$\theta = 32$ rad

$\omega_f = \omega_i + \alpha t$

$\omega_f = 4 \times 4$

$= 16$ rad/s

5) $\tan\theta = \frac{v^2}{rg}$

6)

When the block is moving over wedge, let their common horizontal velocity be v
By the momentum conservation

$mv_0 = (m + M)v$ ..(i)

and from COME

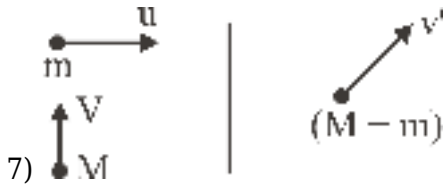
$\frac{1}{2}mv_0^2 = \frac{1}{2}(m + M)v^2 + mgh$ ..(ii)

$\frac{1}{2}mv_0^2 = \frac{1}{2}(m+M)\left(\frac{mv_0}{m+M}\right)^2 + mgh$

$v_0^2\left(1 - \frac{m}{m+M}\right) = 2gh$

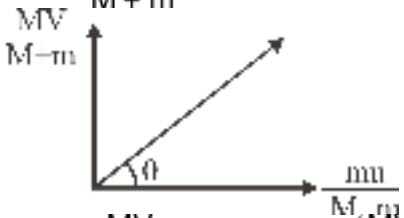
$\Rightarrow v_0 = \sqrt{\frac{2gh(m+M)}{M}}$

7)



$mu\hat{i} + MV\hat{j} = (M + m)v'$

$v' = \frac{mu\hat{i} + MV\hat{j}}{M+m}$



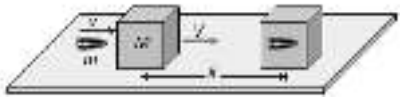
$\tan\theta = \frac{MV}{mu} \Rightarrow \theta = \tan^{-1}\left(\frac{MV}{mu}\right)$

$V' = \sqrt{\frac{(mu)^2 + (MV)^2}{(M+m)^2}}$

$P' = (M + m)v'$

$= \sqrt{(mu)^2 + (MV)^2}$

KE loss $= \frac{1}{2}\left(\frac{Mm}{M+m}\right)(V^2 + u^2)$

8) 

Let speed of the bullet = v
Speed of the system after the collision = V
By conservation of momentum mv = (m + M)V

$$\Rightarrow V = \frac{mv}{M+m}$$

So the initial K.E. acquired by the system

$$= \frac{1}{2}(M+m)V^2 = \frac{1}{2}(m+M)\left(\frac{mv}{M+m}\right)^2 = \frac{1}{2}\frac{m^2v^2}{(m+M)}$$

This kinetic energy goes against friction work done by friction = $\mu R \times x = \mu(m+M)g \times x$
By the law of conservation of energy

$$\frac{1}{2}\frac{m^2v^2}{(m+M)} = \mu(m+M)g \times x$$

$$\Rightarrow v^2 = 2\mu gx\left(\frac{m+M}{m}\right)^2$$

$$\therefore v = \sqrt{2\mu gx}\left(\frac{M+m}{m}\right)$$

9) Let distance of cart and mass is x_c and x_m

$$Mx_c = \frac{M}{2}x_m$$

$$x_c = \frac{x_m}{2}$$

$$x_c + x_m = 10$$

$$\frac{x_m}{2} + x_m = 10$$

$$x_m = \frac{20}{3}\ m$$

10)

Assertion is not correct After the explosion the fragments shoot off in different directions. What does continue on the original path is only the centre of mass of the system not each fragments.
Reason(R) : The explosion occurs due to internal forces only and no external force acts during the explosion.

11) Rate of flow of water through a capillary tube is

$$V = \frac{\pi Pr^4}{8\eta l}$$

As P, η remain the same

$$\therefore \frac{V'}{V} = \frac{(2r)^4}{(r)^4} \times \frac{(l)}{(2l)} = \frac{16}{2} = 8 \Rightarrow V' = 8V$$

12) Let ρ_S, ρ_L be the density of silver and liquid. Also m and V be the mass and volume of silver block.

 Tension in string = mg – bouyant force

$T = \rho_S Vg - \rho_L Vg = (\rho_S - \rho_L) Vg$

Also $V = \frac{m}{\rho_s}$

$\therefore T = \left(\frac{\rho_S - \rho_L}{\rho_s}\right) mg = \frac{(10 - 0.75) \times 10^3}{10 \times 10^3} \times 4 \times 10$

$= 37$ N

13) $mg = 2TL \Rightarrow \pi r^2 Ldg = 2TL \Rightarrow \Rightarrow \pi r^2 dg = 2T.$

This relation is independent of L.

14)

$$w_2 - w_1 = \left[\frac{1}{2}(50)(1.02) - \frac{1}{2} \times 30(0.61)\right] \times 10^{-3}$$

$$= \frac{10^{-2}}{2}[5.10 - 1.83]$$

$$= \frac{3.27}{2} \times 10^{-2}$$

$$= 1.68 \times 10^{-2}\ J$$

$$= 16.8 \times 10^{-3}\ J$$

15)

v = vol. of cavities.

V = vol. of solid iron

$$\therefore \quad V = \frac{m_{iron}}{\rho_{iron}} = \frac{W}{\rho_i g} \qquad ...(1)$$

In the water

$\Rightarrow$ decrease in weight = up thrust

$\Rightarrow W - W_0 = (V + v)\rho_w g$

$$\Rightarrow W - W^0 = \frac{W}{\rho_i g}\rho_w g + \rho_w vg$$

$$\Rightarrow W - W_0 - \frac{W\rho_w}{\rho_i} = \rho_w vg$$

$$\Rightarrow W\left[1 - \frac{\rho_w}{\rho_i}\right] - W_0 = \rho_w vg$$

$$\Rightarrow v = \frac{W}{g}\left[\frac{1}{\rho_w} - \frac{1}{\rho_i}\right] - \frac{W_0}{g\rho_w}$$

16) $Q = A_1A_2\sqrt{\frac{2gh}{A_1^2 - A_2^2}}$

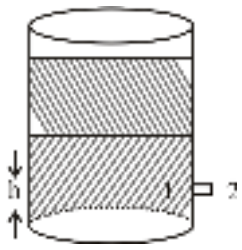
$$= \pi r_1^2 \times \pi r_2^2 \sqrt{\frac{2 \times 10 \times 0.8}{(\pi r_1^2)^2 - (\pi r_2^2)^2}}$$

$$= \frac{\pi^2 r_1^2 r_2^2}{\pi} \sqrt{\frac{16}{(225 \times 10^{-4})^2 - (25 \times 10^{-4})^2}}$$

$$= \frac{10 \times 225 \times 10^{-4} \times 25 \times 10^{-4} \times 4}{\pi \times 10^{-4} \sqrt{250 \times 200}}$$

$$= \frac{225 \times 25 \times 4 \times 10^{-3}}{\pi \times 10^2 \sqrt{5}}$$

$$= \frac{45\sqrt{5}}{\pi} \times 10^{-3} \text{ m}^3/\text{s} = 32 \text{ ltr/s}$$

17) 

Applying Bernoulli theorem between points 1 and 2

$$P^0 + \frac{H}{2}dg + \left(\frac{H}{2} - h\right) 2dg = P_0 + \frac{1}{2}(2d)v^2$$

Solving $v = \sqrt{\frac{(3H - 4h)g}{2}}$

18) $|\overrightarrow{\Delta P}_x| = mv \sin 60° = \frac{\sqrt{3}}{2} mv$

$$|\overrightarrow{\Delta P}_y| = \frac{mv}{2} + mv = \frac{3}{2} mv$$

$$\Rightarrow |\overrightarrow{\Delta P}_{net}| = \sqrt{\Delta P_x^2 + \Delta P_y^2} = \sqrt{\left(\frac{9}{4} + \frac{3}{4}\right)} mv$$

$$|\overrightarrow{\Delta P}_{net}| = \sqrt{3}mv$$

$$\Rightarrow |\overrightarrow{\Delta F}_{net}| = \sqrt{3}\left(\frac{dm}{dt}\right).v = \sqrt{3}\rho Av^2$$ **Ans.** $\left(\text{Since, } dm = A(vdt)\rho \Rightarrow \frac{dm}{dt} = A\rho v\right)$

19) For level flight

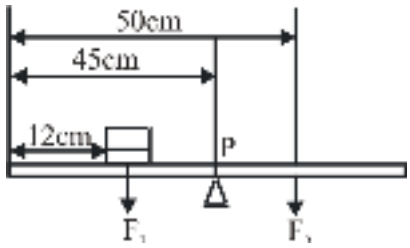
$$Mg = \Delta P \times A = \frac{1}{2}\rho_{air}\left(V_1^2 - V_2^2\right) A$$

$$M \times 10 = \frac{1}{2} \times 1 \times [(75)^2 - (65)^2] \times (2 \times 25)$$

$M = 3500$ kg

20) $\tau = I\alpha \Rightarrow \propto = \frac{\tau}{I}$

$\alpha = \dfrac{(F \sin 30^\circ) L}{\frac{ML^2}{3}} = \dfrac{3F}{2ML}$ Hence option (1)

21) 

"A metre stick is balanced on a knife edge at its centre" means its centre of mass is at its centre.

Let m is the required mass of stick in grammes

$F_1 = (5 + 5) \times 981$ dynes

$F_2 = m \times 981$ dynes

By taking torque about point P

$F_1 \times (45 - 12) = F_2 \times (50 - 45)$

$\Rightarrow 33F_1 = 5F_2$

$33 \times 10 \times 981 = 5 \times (m \times 981)$ so m = 66 gm.

22) $\because \vec{L}_P = \vec{L}_{cm} + \vec{r} \times \vec{p}_{cm} = I_{cm}\vec{\omega} + \vec{R} \times m\vec{v}_{cm}$;

here $v_{cm} = v_0$

Since sphere is in pure rolling motion hence,

$\omega = v_0/R$

$\Rightarrow \vec{L}_p = \left(\frac{2}{5}MR^2\frac{v_0}{R}\right)\left(-\hat{k}\right) + Mv^0R\left(-\hat{k}\right)$

$= \frac{7}{5}Mv^0R\left(-\hat{k}\right)$

23)

Both assertion and reason are true and reason is the correct explanation of assertion.

$\text{Total KE} = \frac{1}{2}m\nu^2 + \frac{1}{2}I\omega^2$

$= \frac{1}{2}m\nu^2 + \frac{1}{2}mK^2\omega^2 = \frac{1}{2}m\omega^2(R^2 + K^2)$

KE of rotation $= \frac{1}{2}I\omega^2\frac{1}{2}(mK^2)\omega^2$

$\therefore \quad \dfrac{\text{KE of rotation}}{\text{Total KE}} = \dfrac{K^2}{R^2 + K^2}$

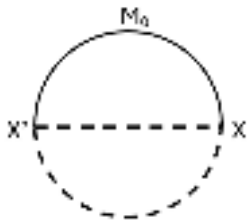
24) For rolling upon the incline plane the retardation

$a = \dfrac{g \sin\theta}{1 + \frac{K^2}{R^2}} = \dfrac{10\ (\sin 30^\circ)}{1 + \frac{2}{5}} = 25/7\ m/s^2$

so distance covered $= \dfrac{v^2}{2a} = \dfrac{(2)^2}{2 \times 25/7}$

= 0.56 m = 56 cm

25)

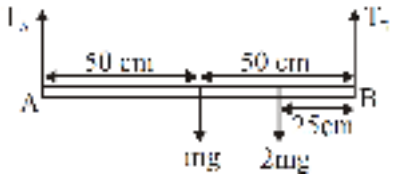


$L = \pi R \Rightarrow R = \frac{L}{\pi}$

For the complete ring, M.I. of the ring about diameter $= \frac{1}{2}(2M_0)R^2$. For the semi-circular ring.

$M.I. = \frac{1}{2}\left[\frac{1}{2}(2M_0)R^2\right]$

$= \frac{1}{2}M_0R^2 = \frac{1}{2}M_0\left(\frac{L}{\pi}\right)^2 = \frac{M_0L^2}{2\pi^2}$



26) $\tau_B = 0$ (torque about point B is zero)

$(T_A) \times 100 - (mg) \times 50 - (2mg) \times 25 = 0$

$100\ T_A = 100\ mg$

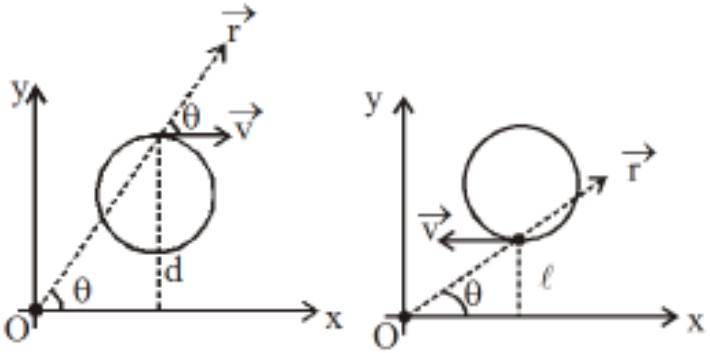
$T_A = 1\ mg$

27)

$a_1 = R\alpha - a_2 \Rightarrow 2 = 2\alpha - 4$

$\alpha = 3\ rad/s^2$

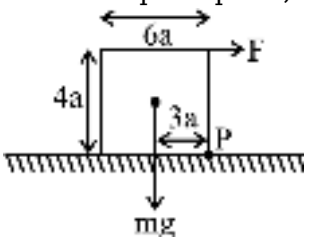
28) As particle moves with constant speed but the magnitude & direction of angular momentum changes continuously. As shown in figure because perpendicular distance of the particle and changes continuously.



$\vec{L} = mvd(-\hat{k})$ $\vec{L} = mv\ell(\hat{k})$

29)

An object begins to topple when the torque from the applied force equals or exceeds the restoring torque from its own weight. We can find this minimum force by equating the torques about the pivot point, which is the bottom corner of the block opposite to the force.



F_{min} for toppling (Torque about point P)

F_{min} (4a) = mg(3a)

$F_{min} = \frac{3mg}{4}$

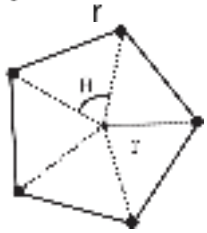
30)

$T^2 \propto R^3$

as radius of orbit increases then, time period also increases.

31) $I = \frac{Gm}{r^2}$

$V = \frac{-Gm}{r}$



Gravitational field at center is zero : (A - R)

Gravitational potential at center = $\left(-\frac{Gm}{r}\right) \times 5$ (B - S)

When one mass is removed field strength with 4 mass $I = \frac{Gm}{r^2}$ (C - P)

Potential with 4 mass $V = \left(-\frac{Gm}{r}\right) \times 4 = -\frac{4Gm}{r}$

$V = \left(-\frac{Gm}{r}\right) \times 4 = -\frac{4Gm}{r}$ (D - S)

32)

Given $V = 5x^2y$

$E_x = \frac{-\partial V}{\partial x} = -10xy$; $E_y = \frac{-\partial V}{\partial y} = -5x^2$

at point (2, 3) ; $\vec{E} = E_x\hat{i} + E_y\hat{j}$

$\vec{E} = -60\hat{i} - 20\hat{j}$

33) Kinetic energy = Potential energy

$$\frac{1}{2}m(kv_e)^2 = \frac{mgh}{1+\frac{h}{R}} \Rightarrow \frac{1}{2}mk^2\,2gR = \frac{mgh}{1+\frac{h}{R}}$$

$$\Rightarrow h = \frac{Rk^2}{1-k^2}$$

Height of Projectile from the earth's surface = h

Height from the centre $r = R + h = R + \frac{Rk^2}{1-k^2}$

By solving $r = \frac{R}{1-k^2}$

34) $p = mv_0 = m\sqrt{\frac{GM}{r}}$ i.e. $p \propto 1/\sqrt{r}$

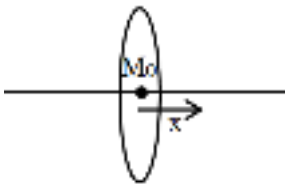
$$\frac{p_A}{p_B} = \sqrt{\frac{r_B}{r_A}} = \sqrt{\frac{R+3R}{R+R}} = \sqrt{2}$$

35) $V^0 = \sqrt{\frac{GM}{r}}$

$V = \sqrt{\frac{GM}{R}}$

$V' = \sqrt{\frac{GM}{4R}} = \frac{V}{2}$

36)



$$F = -M_0\frac{GMx}{(R^2+x^2)^{3/2}} \approx -\frac{GMM_0x}{R^3}$$

$= - M_0\omega^2x$

$\omega = \sqrt{\frac{GM}{R^3}} \Rightarrow T = 2\pi\sqrt{\frac{R^3}{GM}}$

$t = \frac{T}{4} = \frac{\pi}{2}\sqrt{\frac{R^3}{GM}}$

37)

For a satellite revolving around earth

$K = -\frac{U}{2} = -E$

$x = \frac{K-2K-K}{K-(-2K)} = -\frac{2}{3}$

$y = \frac{K-(-2K)+K}{-K-K} = -2$

$\therefore \frac{y}{x} = \frac{3}{1}$

38) diameter $d_1 = 3.0 \times 10^{-3}$m , $d_2 = 6.0 \times 10^{-3}$ m

In one limb $h_1 = \frac{4T}{d_1 \rho g}$ and in other limb $h_2 = \frac{4T}{d_2 \rho g}$

$$\Delta h = h_1 - h_2 = \frac{4T}{\rho g}\left[\frac{1}{d_1} - \frac{1}{d_2}\right]$$

$$= \frac{(4)\left(72 \times 10^{-3}\right)}{10^3 \times 10}\left[\frac{1}{3} - \frac{1}{6}\right] \times 10^3$$

$$\Delta h = \frac{(4)(72)}{6} \times 10^{-4}\text{m}$$

$= 4.8 \times 10^{-3}$m = 4.8mm

39) **For concave meniscus :** Adhesive forces > Cohesive forces
For convex meniscus : Cohesive forces > Adhesive forces
Capillary action : Due to both adhesive and cohesive forces
When a larger drops spits : Work is done to increase the surface area, and energy for it comes at the cost of loss of heat
Reason surface tension : Only cohesive forces

40) $F = \eta A\frac{V-0}{y} + 4\eta A\frac{V-0}{d-y}$

$$\frac{dF}{dy} = 0$$

$$-\frac{\eta}{y^2} + \frac{4\eta}{(d-y)^2} = 0$$

$$\frac{d-y}{y} = 2$$

$$y = \frac{d}{3}, \Rightarrow d_1 = \frac{d}{3}, d_2 = \frac{2d}{3}$$

$$\frac{d_2}{d_1} = 2$$

∴ (B)

41) $h = \frac{2T\cos\theta}{rdg}$

$$\therefore \frac{h_2}{h_1} = \frac{T_2}{T_1} \times \frac{\cos\theta_2}{\cos\theta_1} \times \frac{d_1}{d_2} \times \frac{r_1}{r_2}$$

$$\frac{h_2}{h_1} = \frac{140}{70} \times \frac{\cos 60^\circ}{\cos 0^\circ} \times \frac{1}{2} \times 1 = \frac{1}{2}$$

$$\Rightarrow h_2 = \frac{h_1}{2} = 3\text{cm}$$

42) Relative to liquid, the velocity of sphere is $2v_0$ upwards.
∴ viscous force on sphere = $6 \pi \eta r 2v_0$ downward

$= 12 \pi \eta r v_0$ downward

43) Upward force by capillary tube on top surface of liquid is
$f_{up} = 4\sigma a \cos\theta$
If liquid is raised to a height h then we use
$4\sigma a \cos\theta = ha^2 \rho g$

$$\text{or } h = \frac{4\sigma \cos\theta}{a\rho g}$$

44)

F = 2T (6a) + mg
F = 12 aT + mg

45)

This question is based on $Y = \frac{F \cdot L}{A \cdot \Delta L}$.

$$\Delta\ell = \frac{F \cdot L}{A \cdot Y} \Rightarrow \frac{\Delta\ell_S}{\Delta\ell_B} = \frac{F_1 L_1 Y_2}{F_2 L_2 Y_1}$$

$$\frac{\Delta\ell_S}{\Delta\ell_B} = \frac{100\text{N} \times 1.5\text{m} \times 1.0 \times 10^{11}\text{Pa}}{60\text{N} \times 1.0\text{m} \times 2.0 \times 10^{11}\text{Pa}}$$

= 1.25

CHEMISTRY

46)

Explain the question: Find the region of highest electron density for a 3*p* orbital (=1).
Concept: Shape & Orientation of p-orbitals.
Solution:

A. Identify: The quantum number = 1 defines a p-orbital.
B. Geometry: Unlike d-orbitals which often lie between axes, p-orbitals are dumbbell shaped with lobes oriented along a single axis (x,y, or z).
C. Conclusion: The electron is most likely found along that specific axis.

47)

Ion	Electronic configuration	No. of unpaired electron
Co^{+2}	[Ar] $4s^0 3d^7$	3
Cr^{+3}	[Ar] $4s^0 3d^3$	3
V^{+3}	[Ar] $4s^0 3d^2$	2
Mn^{+2}	[Ar] $4s^0 3d^5$	5
Fe^{+3}	[Ar] $4s^0 3d^5$	5
Cr^{+}	[Ar] $4s^0 3d^5$	5
Ni^{+2}	[Ar] $4s^0 3d^8$	2
Fe^{+2}	[Ar] $4s^0 3d^6$	4

ions, having same no. of unpaired e^-, have same magnetic moment.

48) $\frac{1}{\lambda} = R_H z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

for γ-line of Paschen series of Li^{2+}

$z = 3, n_1 = 3, n_2 = 6$

$$\frac{1}{\lambda_{Li^{2+}}} = R_H(3)^2 \left(\frac{1}{3^2} - \frac{1}{6^2}\right) = 9R_H \times \frac{3}{36}$$

$$\lambda_{Li^{2+}} = \frac{36}{27 R_H} = \frac{4}{3R_H}$$

For β-line of Balmer series of He^+

$z = 2, n_1 = 2, n_2 = 4$

$$\frac{1}{\lambda_{He^+}} = R_H(2)^2 \left(\frac{1}{2^2} - \frac{1}{4^2}\right) = 4R_H \times \frac{3}{16}$$

$$\lambda_{He^+} = \frac{4}{3R_H}$$

$$\frac{\lambda_{Li^{2+}}}{\lambda_{He^+}} = 1$$

49) $\lambda \propto \frac{1}{\sqrt{v}}$ for same particle

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{v_2}{v_1}} = \sqrt{4} = 2$$

50)

The two orbits are either I and II or II and IV

$\frac{r_{n_2}}{r_{n_1}} = \frac{4}{1}$ and $r_n \propto n^2$

Thus, $E_2 - E_1 = \frac{-13.6}{4} + 13.6 = 10.2$ eV

and $E_4 - E_2 = \frac{-13.6}{16} + \frac{13.6}{4} = 2.55$ eV

51) For higher energy level : $l = 2$
and $n - l - 1 = 2 \Rightarrow n = 5$
For lower energy level : $l = 0$
and $n - l - 1 = 0 \Rightarrow n = 1$
Now, $326.4 = 13.6 \times z^2 \left(\frac{1}{1^2} - \frac{1}{5^2}\right) \Rightarrow Z = 5$

52)

Isotones have same number of neutrons
17 – 9 = B – 8
B = 16
Isobars have same mass number
A = B = 16
No of neutrons = 16 – 8 = 8

53) Energy of one photon = $\frac{hc}{\lambda}$
Lets assume 'n' no. of photons are present :
$\therefore n \times \frac{hc}{\lambda} = 3$ J
$\Rightarrow n \times \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{662.6 \times 10^{-9}} = 3$
$\therefore n = 10^{19}$

54)

Distance of closest approach is given by
$r = \frac{2k(z_1e)(z_2e)}{mv^2}$
For $r_{minimum}$, z should be minimum

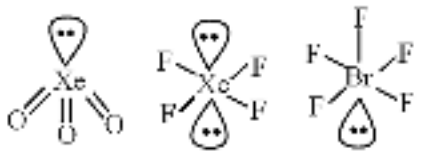
55)

$\Delta x \cdot \Delta p = \frac{h}{4\pi} \quad \left(\Delta x = \frac{h}{p}\right)$
$\frac{h}{p} \Delta p = \frac{h}{4\pi} \Rightarrow \frac{\Delta p}{p} = \frac{1}{4\pi}$
$\Rightarrow \frac{M \cdot \Delta v}{M \cdot v} = \frac{1}{4\pi}$
$\frac{\Delta v}{v} = 8\%$

56)

Covalent compound do not show in liquid state.

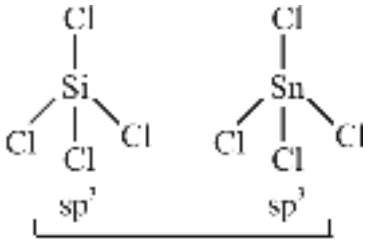
57) $\underset{sp^2}{CH_3} \rightarrow \underset{sp^3}{C_2H_6}$



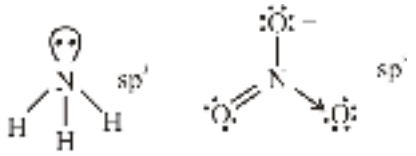
58) $(BA \approx 109.5^a) > (BA = 90^e) > (BA < 90^e)$

59)

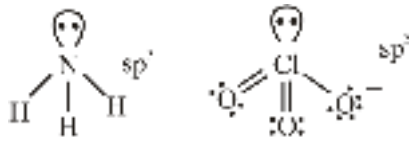
Ionic character $\propto \frac{1}{\oplus ve \text{ charge}}$



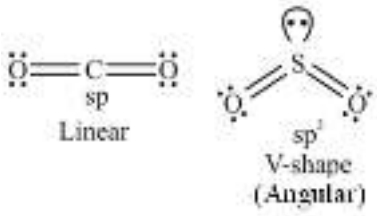
60) (i) Tetrahedral



(ii) Pyramidal Trigonal planar



(iii) Pyramidal Pyramidal



(iv)

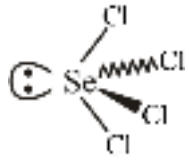
61)

Bond order → $ClO_4^- > SO_4^{2-} > PO_4^{3-}$

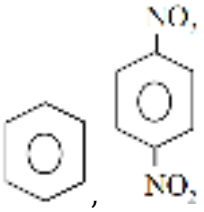
(7/4) (6/4) (5/4)

62)

M.P. → KF > KCl

63) 

sp^3d see-saw shape
1 L.P. + 4 b.p.

64) 

, XeF_4, PCl_3F_2, N_2O_4 are non polar

65)

B.P ∝ no. of H-bond.

66)

If $\vec{u}$ (net) = 0
than dipole moment = 0

67) Compound → Types of B.L.
H_2O → 1
$Cr_2O_7^{2-}$ → 2
SF_6 → 1
PCl_5 → 2

68)

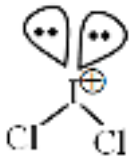
Solution of ionic compounds show conductance due to ions.

69)

$XeOF_4$ – sp^3d^2 - pπ-dπ

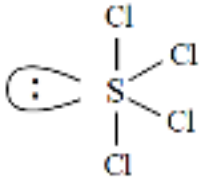
70)

MP → $F_2 < I_2$
(MP ∝ Mass)

71)  (bp + □p) (2 + 2) = 4

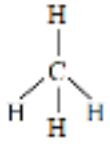
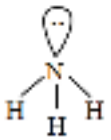
 (2 +3) = 5

 (4 + 2) = 6

72) 

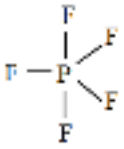
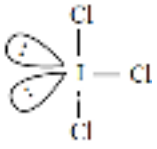
sp^3d, see-saw

73)

(P)  

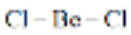
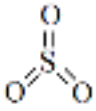
sp^3 sp^3

BA = 109.5° BA < 109.5°

(Q)  

sp^3d sp^3d

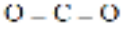
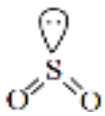
90°, 120° < 90°

(R)  

sp sp^2

180° 120°

planar planar

(S)  

sp sp^2

180° < 120°

planar planar

74)

$XeO_2F_2 \rightarrow sp^3d$, see saw

75)

The correct answer is option (B)
Or
BL ∝ size of atom

76) **Explanation:**
The question asks which of the provided pairs of elements share the same number of electrons in their outermost (valence) electron shell.
Concept: (Electron sharing in their outermost shell)
Selenium (Se) and Tellurium (Te) are both in Group 16 of the periodic table. Elements in Group 16 possess 6 valence electrons. Therefore, Se and Te have the same number of valence electrons.
Final Answer : Option (2)

77) **Question Explanation:** Determine the chemical formula for the halide of element with Z=56.
Concept: Valency and Formula Writing
Solution: The element with atomic number 56 is Barium (Ba), which is in Group 2 and forms a +2 ion (E^{2+}). Halides (X) are in Group 17 and form -1 ions (X^-). To create a neutral compound, two halide ions are needed, making the formula EX_2.
Final Answer: Option (3)

78)

IP. order
$(n-1)d^{10} > np^1$
Zn > Ga
Cd > In
Hg > T□

79)

It is a f-block elements

80)

(A) Z = 37 → s-block
(B) Z = 42 → d-block
(C) Z = 34 → p-block
(D) Z = 92 → f-block

81)

Cu < Zn < Ca < K
L → R Size↓ but ($r_{Cu} < r_{Zn}$)

82) NCERT - Part-I (Class XI); Table 3.7, Page No. 07

83)

L → R Z_{eff} ↑ Size↓
$Yb^{+3} < Pm^{+3} < Ce^{+3} < La^{+3}$

84)

EN order
S < Cl < O < F
Size order
F < O < N < Na

85)

B > Al < Ga > In < T□
IE ⇒ Al < Ga (due to transition contraction)
In < T□ (due to lanthanide contraction)

86)

(1) HF < HCl < HBr < HI (Acidic nature)
(2) HOCl > HOBr > HOI (Acidic nature)
(3) NaOH > $Mg(OH)_2$ (Basic nature)

87) C = 2.5
N = 3
Si = 1.8
P = 2.1

88)

EA order
Cl > O > C > N

89)

EA order Cl > F > Br > I
S > Se > Te > O

90)

EN order C < N
N → $2s^2$, $2p^3$
EN does not depend on stable configuration

BIOLOGY

91) NCERT Pg.# English- 95,96,98,100.
Hindi- 96,97,98,100.

92) NCERT Class-XI : Page No. 96-101

93) NCERT Pg.# 96,100,102.

94) NCERT, Pg # 98

95) NCERT Pg. No. 95, 96

96) NCERT Pg. No. 88

97) NCERT Pg. # 101

98) NCERT Pg. # 96

99) NCERT Pg. No. # 95

100) NCERT-XI, Pg # 94

101) NCERT-XI, Pg # 96

102) NCERT Page No. 85

103) NCERT Pg. No. 98

104) Module Pg.# 100

105) NCERT Pg. # 97

106) XI-NCERT Page No. # 125

107) NCERT Pg. # 123, 124

108) NCERT, Pg # 124

109) NCERT-XI, Pg. # 123, 124

110) NCERT Pg. # 127

111) Module Pg.# 196

112) NCERT Pg # 121

113) Module

114) NCERT, Pg. # 127,128

115) NCERT Pg # 124

116) NCERT-XI, Pg. # (E)-122, (H)-122

117) NCERT, Pg. # 128

118) NCERT-XI Pg. # 123, 124, 127

119) NCERT-XI Pg. # 121, 122, 124

120) XI-NCERT Page No. # 126

121)

NCERT-XI, Pg. # 110

122)

NCERT-XI, Pg. # 116

123) NCERT XI Pg No. # 118

124) NCERT Pg. # 116,117

125) NCERT Pg.#106.

126) NCERT Pg.#106, 109, 110.

127) NCERT, Pg. # 110

128)

NCERT Pg. # 106

129)

NCERT Pg. # 112

130) NCERT, Pg. # 109

131) NCERT Page No. 110, 109

132) NCERT, Pg. # 110

133) NCERT XI Pg No. # 117

134) NCERT XI Pg. No. # 117

135) NCERT Pg.# No. 115

136)

NCERT XI Pg. # 189

137)

NCERT XI Pg. # 186

138) NCERT Pg. No. # 195, 195

139)

NCERT XI Pg. # 200

140)

NCERT XI Pg. # 202

141)

NCERT XI, Pg. # 196

142) NCERT Pg # 212

143)

NCERT XI, Pg. # 205,206

144)

NCERT XI, Pg. # 212

145)

NCERT _ Pg. # 184

146)

NCERT Pg#184

147) NCERT Pg#183

148) NCERT-XI, Pg. # 270-71

149)

NCERT Pg. # 189-190

150) NCERT Pg # 189, 190

151)

NCERT Pg. # 202

152)

NCERT Pg. # 199

153) NCERT-XI, Pg. # 279-280

154) NCERT-XI, Pg. # 280-81

155)

NCERT Pg. # 200

156) NCERT Pg # 206, 207, 208

157) NCERT Pg # 209

158) NCERT Pg # 210

159) NCERT Pg. # 214

160)

NCERT Pg. # 209

161) NCERT_Pg. # 190-191

162)

NCERT Pg. # 184

163)

NCERT Pg. # 191

164)

NCERT_ Pg. # 187

165)

NCERT Pg. # 184

166) NCERT Pg. # 187

167)

NCERT-XI, Pg. # 270

168) NCERT Pg. # 194

169) NCERT Pg # 200

170) NCERT Pg. # 197

171)

NCERT Pg. No. # 197

172)

NCERT Pg. # 201

173) NCERT Pg. No. # 287

174) NCERT - Pg#284(E), 285 (H)

175) NCERT Pg # 207

176) NCERT Pg.# 213

177) NCERT-XI, Pg # 298

178) NCERT XI Pg. # 211

179)

NCERT Pg # 294

180) NCERT, Pg. # 204, Summary